

Stablecoins and the Future of Payments: Evidence from Financial Markets

Alexander Copestake, Cage Englander,
Maria Soledad Martinez Peria, Germán Villegas-Bauer

WP/26/52

IMF Working Papers describe research in progress by the author(s) and are published to elicit comments and to encourage debate.

The views expressed in IMF Working Papers are those of the author(s) and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

**2026
MAR**



IMF Working Paper
Research Department

**Stablecoins and the Future of Payments:
Evidence from Financial Markets**

**Prepared by Alexander Copestake, Cage Englander,
Maria Soledad Martinez Peria, Germán Villegas-Bauer***

Authorized for distribution by Giovanni Dell'Ariccia

March 2026

IMF Working Papers describe research in progress by the author(s) and are published to elicit comments and to encourage debate. The views expressed in IMF Working Papers are those of the author(s) and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

ABSTRACT: We examine whether financial market participants, in aggregate, expect stablecoins to play an important role in payments. Using high-frequency variation in stock prices, we estimate that U.S. legislation supporting the use of stablecoins in payments reduced the market value of listed incumbent payment firms by 18% or approximately \$300 billion, consistent with stablecoins increasing competition in the payments sector. This impact is larger than that of other recent pro-competitive regulatory shocks and (i) proportionately larger for incumbents focused on cross-border payments, (ii) smaller for incumbents protected by network effects, and (iii) smaller for incumbents already offering crypto-related services.

JEL Classification Numbers:	E42, G12, G23, G28
Keywords:	Crypto Assets, Stablecoins, Payments, Stock Returns
Author's E-Mail Address:	acopestake@imf.org , cenlander@imf.org , mmartinezperia@imf.org , gvillegasbauer@imf.org

* We thank Itai Agur, Jennie Bai, Damien Capelle, Thomas Drechsel, Adriano Fernandes, Melih Firat, Janet Gao, Maximilian Grimm, Martina Hengge, Diego Känzig, Divya Kirti, Thomas Kroen, Samantha Myers, Marco Reuter, Andrew Usher, and various seminar participants at the IMF for helpful comments and suggestions.

What is needed is an electronic payment system based on cryptographic proof instead of trust, allowing any two willing parties to transact directly with each other without the need for a trusted third party. [Nakamoto \(2008\)](#)

Ever since crypto's invention, enthusiasts have promised that blockchain tokens will find widespread legal use cases, displacing conventional means of payment, any day now. But it keeps not happening. [Krugman \(2025\)](#)

1 Introduction

Will crypto assets play an important role in payments? While [Nakamoto \(2008\)](#) aimed to provide an electronic version of cash, usage of bitcoin for payments remains concentrated in illicit transactions (e.g., [Rogoff, 2025a,b](#)).¹ Recently, proponents of stablecoins have predicted mass adoption in payments (e.g., [Allaire, 2025](#)), yet skeptics still consider crime to be the primary function of crypto assets (e.g., [Krugman, 2025](#)).²

Empirically assessing the potential role of stablecoins in payments is challenging. First, adoption of new payments technologies can be slow initially because potential users wait for a critical mass of others to adopt first ([Alvarez, Argente, Lippi, Méndez, and Van Patten, 2023](#)). Second, fewer than 10% of stablecoin transactions recorded on blockchains reflect transactions between genuine users ([Allium Labs, 2025](#)).³ Third, blockchain data alone cannot reveal whether such transactions are payments for goods and services (rather than, for example, investments) because the corresponding flows of goods and services are not recorded.

In this paper, we shed light on the potential role of stablecoins in payments through an indirect approach: examining financial markets' valuations of incumbent payment firms. Financial market reactions to technology shocks have a history of predictive power (e.g., [Kogan, Papanikolaou,](#)

¹Nakamoto, in turn, built upon a series of prior designs for digital equivalents of cash. See, for example, “DigiCash” (which evolved from [Chaum, 1983](#)), “b-money” ([Dai, 1998](#)), “Hashcash” ([Back, 2002](#)), “reusable proofs-of-work” ([Finney, 2004](#)), and “bit gold” ([Szabo, 2005](#)).

²Stablecoins are crypto assets that aim to maintain a stable value relative to a specified asset or a pool or basket of assets ([Adrian et al., 2025](#)).

³The remainder reflect, for instance, activity by bots or re-balancing between different wallets owned by the same centralized exchange.

[Seru, and Stoffman, 2017](#)). Investors in incumbent payment firms must take a stand on the future landscape within which those firms will operate, and their reactions to news about stablecoins reflect these expectations.⁴

We study the stock market valuations of U.S.-listed payment firms around key congressional votes in the passage of [S.1582](#), the bill that would become the Guiding and Establishing National Innovation for U.S. Stablecoins Act (the GENIUS Act). This Act reduces regulatory uncertainty by creating the first U.S. federal regulatory system for “payment stablecoins,” and increases trust in the stability of such stablecoins by requiring 100% backing with liquid assets and monthly public disclosures of reserves ([The White House, 2025](#)).⁵ Importantly, the GENIUS Act passed with bipartisan support, suggesting a persistent change in the regulatory environment for stablecoin issuers.

In our baseline specification, we focus on the ten trading hours surrounding the decisive congressional vote that sent the bill to the president to be signed into law. We compare the stock returns of incumbent payment firms to those of other financial firms, controlling for company and time fixed effects. In the five hours prior to the vote, the two sets of stock prices follow a similar path. Then, the two sets of prices diverge: on average, incumbent payment firms’ stock returns are approximately 1% lower than those of other financial firms in the five trading hours after the vote. These differences in returns are statistically significant at the 1% level and equate to approximately \$21.5 billion of lost market capitalization for incumbent payment firms.

Markets had already partially anticipated Congress passing the GENIUS Act prior to the vote. As a result, the immediate price response did not reflect the full effect of the GENIUS Act becoming law. In a second step, we therefore exploit data from prediction markets on the expected probability of such a bill being enacted, prior to the decisive vote. This allows us to gauge the

⁴Indeed, recent evidence using announcements related to central bank digital currencies ([Burlon, Muñoz, and Smets, 2024](#); [Berg, Keil, Martini, and Puri, 2024](#)) suggests that markets pay close attention to news concerning digital payments technologies.

⁵The GENIUS Act defines a “payment stablecoin” as “a digital asset (i) that is, or is designed to be, used as a means of payment or settlement; and (ii) the issuer of which— (I) is obligated to convert, redeem, or repurchase for a fixed amount of monetary value, not including a digital asset denominated in a fixed amount of monetary value; and (II) represents that such issuer will maintain, or create the reasonable expectation that it will maintain, a stable value relative to the value of a fixed amount of monetary value.”

extent of the dampening effect of anticipation and so reverse-engineer the full “no anticipation” effect of the legislation.⁶ We estimate that the passage of the GENIUS Act in 2025 reduced the total market capitalization of incumbent payment firms by 18%, or approximately \$300 billion.

To unpack the drivers of these differential returns, we consider how the decline in stock returns varies with the characteristics of the incumbent payment firms. We inspect the firms’ regulatory filings and classify them on three important dimensions: (i) whether the firm’s primary business focuses on cross-border rather than domestic payments; (ii) whether proprietary network effects are central to their existing business model; and (iii) whether they offered crypto-related services prior to the decisive vote on S.1582. We find that the differential negative returns after the vote are significantly larger for incumbents focused on cross-border payments, absent for incumbents whose business entails proprietary network effects, and absent for firms that already engaged with crypto assets prior to the vote.

We conclude by benchmarking our results against other historical shocks to payment firms. Our estimated impact of U.S. stablecoin legislation on U.S. payment firms is slightly larger than comparable estimates for other recent pro-competitive regulatory shocks such as the Durbin Amendment and the European Central Bank’s plan to issue a digital euro.

Together, our results suggest that financial market participants expect stablecoins to substantially but heterogeneously increase the competitive pressure faced by incumbent payments providers. Cross-border payments firms are expected to be most exposed, consistent with the starker contrast between relatively slow and expensive existing cross-border payments systems (e.g., [FSB, 2025](#); [Smets, 2025](#)) and the inherently borderless blockchain infrastructure underpinning stablecoins. In contrast, firms with stronger existing network effects are expected to be less exposed—consistent with this source of competitive advantage being less vulnerable to disruption than, for instance, technical expertise in the existing technology stack. Finally, our results suggest that engaging with the new technology could help firms prosper in a world where stablecoins play an increasingly important role in payments.

⁶See [Snowberg, Wolfers, and Zitzewitz \(2011\)](#) and [Hazell and Hobler \(2024\)](#) for similar approaches using prediction markets.

Literature. We contribute to three main strands of literature: on payments, on blockchains and their applications, and on high-frequency identification.

First, a substantial literature examines the adoption of different payments technologies. Building on early models of network effects in payments (e.g., [Weinberg, 1997](#); [Rochet and Tirole, 2003, 2004](#)), recent work has extended and tested the theory in the contexts of debit and credit cards, point-of-sale machines, and mobile payments apps (e.g., [Crouzet, Gupta, and Mezzanotti, 2023](#); [Alvarez, Argente, Lippi, Méndez, and Van Patten, 2023](#); [Higgins, 2024](#); [Wang, 2025](#); [Copestake, Kirti, Martinez Peria, and Zeng, 2025b](#)). Focusing on digital assets, our work relates most closely to [Burlon, Muñoz, and Smets \(2024\)](#) and [Berg, Keil, Martini, and Puri \(2024\)](#), who study potential adoption of a central bank digital currency. We contribute by examining stablecoins and their potential for widespread adoption in payments, building upon [Liao, Hadeed, and Zeng \(2023\)](#), who document the emergence of payment stablecoins. Our work is forward-looking, revealing that financial market participants expect stablecoins to play a significant role in the payments landscape, and highlights the differential exposure of different types of incumbents.

Second, a growing literature in economics and finance examines blockchains and their applications. One sub-strand explores the economic uses of blockchains, with recent theoretical work highlighting both novel capabilities ([Chiu and Koeppl, 2019](#); [Cong and He, 2019](#); [Saleh, 2021](#); [Reuter, 2024](#); [Li and Mann, 2025](#)) and fundamental limitations ([Hinzen, John, and Saleh, 2022](#); [Budish, 2025](#); [Ebrahimi, Guennewig, Routledge, and Zetlin-Jones, 2025](#)).⁷ In this sub-strand, recent empirical papers use historical data from blockchain transactions and/or crypto exchanges to measure stablecoins' convenience yield ([Gorton, Ross, and Ross, 2022](#)) and to assess the use of crypto assets in cross-border payments ([Graf von Luckner, Reinhart, and Rogoff, 2023](#); [Graf von Luckner, Koepke, and Sgherri, 2024](#); [Auer, Lewrick, and Paulick, 2025](#); [Cerutti, Chen, and Hengge, 2025](#)). Our paper complements this work by using stock price data on incumbent payment firms to provide forward-looking empirical evidence on the expected future role of sta-

⁷For a discussion of the uses, risks, and opportunities of tokenization more broadly, see, for example, [Agur, Villegas-Bauer, Mancini-Griffoli, Martinez Peria, and Tan \(2025\)](#).

blecoins in both domestic and cross-border payments.⁸

Another sub-strand of the literature on blockchains examines risks and spillovers that could become important if stablecoins are widely adopted. Several recent papers model stablecoin issuers and the mechanisms that could lead stablecoins to depeg (Bertsch, 2023; d’Avernas, Maurin, and Vandeweyer, 2022; Gorton, Klee, Ross, Ross, and Vardoulakis, 2025; Ma, Zeng, and Zhang, 2025; Li and Mayer, 2025; He, Zhao, and Zhou, 2026). Our results suggest that, despite such possibilities, financial market participants expect (some) stablecoins to prove sufficiently credible to play an important role in payments. In addition, Ahmed and Aldasoro (2025), Ferrari Minesso and Siena (2026) and Cerutti, Firat, Hengge, and Sagawa (2026) find that stablecoin issuance already has detectable spillovers to financial markets—spillovers that, our analysis suggests, will become increasingly important as stablecoins’ usage in payments grows.⁹

Finally, our paper also contributes to the literature on high-frequency identification in empirical macroeconomics and finance.¹⁰ Initially focused on monetary policy announcements, this approach is based on the idea that, within a narrow time-span around an important event, changes in asset prices primarily reflect the event rather than other confounding factors. Our use of legislative shocks most closely parallels the regulatory shocks used in Fringuellotti and Kroen (2024) and Drechsel and Miura (2025). Our approach of focusing primarily on a single, high-powered shock builds on Acemoglu, Johnson, Kermani, Kwak, and Mitton (2016), who examine the announcement of Timothy Geithner as nominee for Treasury Secretary, and Hazell and Hobler (2024), who exploit the Georgia Senate election results in early 2021. To the best of our knowledge, we are the first to use this approach to investigate the implications of stablecoins for payments.

⁸In focusing on the uses of blockchains in *payments*, our work also complements recent work in household finance on crypto assets as investment goods, i.e., stores of value (e.g., Aiello, Baker, Balyuk, Maggio, Johnson, and Kotter, 2023a,b; Weber, Candia, Coibion, and Gorodnichenko, 2023).

⁹Cerutti, Firat, Hengge, and Sagawa (2026) also find evidence of positive spillovers from stablecoin demand to the stock prices of some crypto-engaged payments firms, consistent with our finding that such firms are less vulnerable to being disrupted by regulatory changes that support the use of stablecoins for payments.

¹⁰E.g., Kuttner (2001), Gürkaynak, Sack, and Swanson (2005), Nakamura and Steinsson (2018), Jarociński and Karadi (2020), Känzig (2021), Swanson (2021), and Bauer and Swanson (2023).

Outline. In Section 2 we discuss the possible implications of stablecoins for incumbent payment firms and present hypotheses that can be tested using stock returns. In Section 3 we describe the empirical setting, data, and regression specifications that we use to test our hypotheses. Section 4 presents our results and benchmarks them against other historical shocks affecting payment firms. Section 5 concludes.

2 Conceptual framework

Stablecoins and incumbent payment firms’ cash flows. If stablecoins come to play an important role in payments, there are two main channels through which they could affect the future cash flows of incumbent payment firms.

First, stablecoins could increase competition and erode rents. Public blockchains are global open-access ledgers, so they reduce the up-front infrastructure requirements involved in offering payment services. Thus, new entrants could use stablecoins to offer payment services at scale and end users can make peer-to-peer payments—in both cases, potentially bypassing incumbent payment firms and reducing their profits.

Second, incumbent payment providers could use stablecoins to reduce the marginal cost of providing payments, which—holding competition constant—could increase profits by raising markups and/or expanding demand. Indeed, stablecoins have already become a relatively cheap means of moving money, offering settlement of any volume in less than a second for less than \$0.01 (e.g., [a16z crypto, 2025](#)).

Expectations and market prices. Financial market participants’ expectations of future market power and future costs are reflected in the stock prices of incumbent payment firms (e.g., [Cho, Grotteria, Kremens, and Kung, 2026](#)). Expectations of the impacts of technology shocks, as reflected in prices, have historically been predictive (e.g., [Kogan, Papanikolaou, Seru, and Stoffman, 2017](#)). We therefore study how the stock prices of incumbent payment firms respond to shocks that support the use of stablecoins in payments. Let such a shock occur at time t^* and let the causal

impact of the shock on firm i 's stock price return between time $t^* - 1$ and time t be $\eta_{i,t}$.

We first consider the impact of stablecoins on the average incumbent payment firm. Denote the set of payment firms by $\mathcal{G}^P$ and the average value of $\eta_{i,t}$ within that group by $\bar{\eta}_t(\mathcal{G}^P) = \frac{1}{|\mathcal{G}^P|} \sum_{i \in \mathcal{G}^P} \eta_{i,t}$. Under the null hypothesis that stablecoins are not expected to play an important role in payments, payment firms' stock prices are unaffected by the shock, so $\bar{\eta}_t(\mathcal{G}^P) = 0$. We conjecture that, instead:¹¹

Hypothesis 1. *Financial markets expect stablecoins to play an important role in payments, with the “higher competition” channel dominating on average. A shock that supports the use of stablecoins in payments therefore reduces the market value of incumbent payment firms: $\bar{\eta}_t(\mathcal{G}^P) < 0$.*

Next, we turn to heterogeneity among incumbent payment firms. Compared to domestic payments, cross-border payments are slower and more expensive (FSB, 2025; Smets, 2025). Given that the blockchain infrastructure underpinning stablecoins is inherently borderless, payment firms focused on cross-border payments could be especially exposed to increased competition from stablecoin-based challengers. Denoting by $\mathcal{G}^X$ the set of payment firms whose primary business focuses on cross-border payments, and by $\bar{\eta}_t(\mathcal{G}^X)$ the analogue of $\bar{\eta}_t(\mathcal{G}^P)$ for such firms, we have:

Hypothesis 2. *Cross-border payment firms are expected to be especially exposed to stablecoins playing a greater role in payments. A shock supporting the use of stablecoins in payments therefore has a larger negative impact on cross-border payment firms than on other incumbent payment firms: $\bar{\eta}_t(\mathcal{G}^X) < \bar{\eta}_t(\mathcal{G}^P)$.*

Among payment firms focused on domestic payments, the relative importance of the two channels is likely to vary substantially by business model. Incumbent firms that operate networks, such as Visa or PayPal, benefit from strong direct and indirect network effects: users (whether consumers, merchants, or financial institutions) have stronger incentives to remain in the network when its user base is large. These network effects could protect such firms from increased compe-

¹¹ Stablecoin-native firms have already launched services (e.g., CPN, BVNK) that compete with existing incumbent payment firms, suggesting that stablecoin-driven competition is already materializing.

tion. Denoting by $\mathcal{G}^N$ the set of incumbent payment firms whose primary business is operating one or more networks, we have:

Hypothesis 3. *Firms operating networks are expected to be less exposed to stablecoins playing a greater role in payments. A shock supporting the use of stablecoins in payments therefore has a less negative impact on network operators than on other incumbent payment firms: $\bar{\eta}_t(\mathcal{G}^N) > \bar{\eta}_t(\mathcal{G}^P)$.*

Finally, regardless of their business model, incumbent payment firms that are early to engage with blockchain technology may be better positioned to exploit opportunities that arise or to respond to new competition. Grouping crypto-engaged payment firms into the set $\mathcal{G}^E$, which we define in detail in the next section, our final hypothesis is then:

Hypothesis 4. *Crypto-engaged payment firms are expected to be less vulnerable to stablecoin-driven disruption, so a shock supporting the use of stablecoins in payments has a less negative impact on them than on other incumbent payment firms: $\bar{\eta}_t(\mathcal{G}^E) > \bar{\eta}_t(\mathcal{G}^P)$.*

Before turning to our empirical design, we consider how to interpret these causal impacts (e.g., $\bar{\eta}_t(\mathcal{G}^P)$) in the context of an event that is partially anticipated.

Anticipation and magnitudes. Let the shock at time t^* be the release of information that leads market participants to update the probability that they assign to some policy d being enacted.¹² Denote the probability that market participants assign to d before the shock by p_{t^*-1} and denote the probability that they assign to d at some subsequent time t by p_t . We are interested in the full impact on stock returns of d being enacted, which we denote by η_i^d . Following [Snowberg, Wolfers, and Zitzewitz \(2011\)](#), this can be estimated by

$$\eta_i^d = \frac{\eta_{i,t}}{p_t - p_{t^*-1}} \quad (1)$$

i.e., we can back out the full effect of the policy d by scaling the observed causal impact of the shock ($\eta_{i,t}$) by the size of the probability update ($p_t - p_{t^*-1}$).

¹²In our main specification, described below, the shock at t^* is a specific congressional vote while the policy d is the GENIUS Act.

3 Empirical design

In this section, we first provide context on the GENIUS Act, then we describe our data and regression specifications.

3.1 The GENIUS Act

The conceptual framework above requires a shock that (a) affects the future use of stablecoins in payments, and (b) is sufficiently important that it dominates the background noise in stock price returns ([Casini and McCloskey, 2025](#)).

To achieve these criteria, we examine legislative events relating to Senate Bill 1582 ([S.1582](#)), which would become the Guiding and Establishing National Innovation for U.S. Stablecoins Act (the GENIUS Act). This Act established the first federal framework for the regulation of payment stablecoins in the U.S., reducing regulatory uncertainty and increasing confidence in their use. Key elements include the requirements for 100% reserve backing with liquid assets, for issuers to publish monthly disclosures of these reserves, and for independent audits for issuers exceeding certain size thresholds ([The White House, 2025](#)). Moreover, the law ultimately passed with bipartisan support ([Figure A.1](#)), suggesting that it would have persistent effects rather than being reversed in the near future.

We focus on the ten trading hours surrounding the decisive congressional vote on S.1582, namely its approval by the House of Representatives on July 17, 2025. This vote has several advantages for our purposes. First, it occurred during a period in which financial market participants were already paying heightened attention to crypto-related legislation. Specifically, the vote occurred during what congressional leadership dubbed “[Crypto Week](#),” a concerted push by lawmakers to progress crypto-related legislation that was widely covered by the financial press.¹³ The GENIUS Act was the only piece of crypto-related legislation to become law during “Crypto Week” and attracted by far the most attention ([Figure A.3](#)). Thus, the marginal investor was especially

¹³See, for instance, coverage in [Bloomberg](#), the [Financial Times](#) and [The Wall Street Journal](#).

likely to be aware of S.1582 and to factor it into their trading decisions both before and after it passed the House.

Second, the House of Representatives passing S.1582 resolved the last significant uncertainty about whether the bill would become law. Passage by the House was the final legislative step needed to send the bill to the president’s desk for signing, since the Senate had already approved the bill and the House took up the Senate’s version without amendments. Whether the bill would be signed into law if presented to the president entailed very little uncertainty, since the president had already [publicly endorsed](#) the bill before it reached the House. After the House vote, the bill was indeed immediately presented to the president and signed into law.

Third, the bill’s passage by the House was not certain *ex ante*. Prior to the vote there was both intra-Republican and cross-party disagreement about whether to advance the bill and also about whether to proceed instead with an alternative measure introduced into the House, [H.R.2392](#) (the STABLE Act). Contemporaneous news reports (e.g., [Politico](#), [Decrypt](#)) highlighted last-minute haggling over votes, with the president intervening directly to persuade key holdouts.¹⁴ Figure 1a validates that the bill’s passage by the House was not perfectly anticipated. The vote was accompanied by a sudden increase in the prediction market-implied probability of stablecoin legislation passing before the end of the year, along with an increase in the prices of the native assets of the Ethereum and Solana blockchains—two of the largest platforms used for stablecoins.

In summary, the passage of S.1582 by the House constituted a substantial and persistent shock that supported the use of stablecoins for payments. Next, we exploit this shock and the response of incumbent payment firms’ stock prices to learn about the expected role of stablecoins in payments.

3.2 Data

We use data on (i) stock prices, balance sheets and income statements, (ii) legislative events, (iii) payment firm characteristics, and (iv) prediction market-implied probabilities. Datasets (ii) and

¹⁴First-hand accounts also emphasize this uncertainty. For instance, the White House AI and Crypto Czar stated that: “The legislation was declared dead many times... There are a lot of twists and turns in the legislative process; it’s definitely true that you don’t want to see the sausage getting made” ([Sacks, 2025](#)).

(iii) are constructed for this study and made available with the paper.

Stock prices, balance sheets, and income statements. We use data from Bloomberg on stock prices for three sets of firms: (i) payment firms, defined as those under Global Industry Classification Standard (GICS) code *40201060*, as described further below; (ii) other financial sector firms, specifically those indicated by prefix *40* in GICS except for those in set (i); and (iii) firms in the S&P 500 except for those in sets (i) and (ii). We combine these with balance sheet and income statement data, also from Bloomberg.

Legislative event timestamps. Congress maintains a [timeline](#) of all congressional actions taken during the legislative process, including formal debates, votes, and proposals. Our baseline specification focuses on the approval of S.1582 by the House of Representatives on July 17, 2025. In robustness checks (described below) we also consider a broader set of events. For each event that we consider, we construct minute-level timestamps using [C-SPAN’s congressional vote archive](#). Specifically, we review footage to determine the minute of the decisive vote (i.e., the vote that put the total count over the required threshold).

Payment firms. We classify payment firms as those in [Global Industry Classification Standard](#) (GICS) code *40201060*, “Transaction & Payment Processing Services”.¹⁵ For our sample period, this definition yields 35 listed U.S. payment firms, with a combined market capitalization of \$1.5 trillion in 2024—equivalent to 77% of the aggregate market capitalization of all listed U.S. commercial banks. To test Hypotheses 2-4, we also construct three sub-groups of payment firms based on their regulatory filings. First, we identify firms whose primary business focuses on cross-border payments (set $\mathcal{G}^X$). Second, we identify firms whose primary business involves operating one or more networks (set $\mathcal{G}^N$). Finally, we identify firms that had engaged with crypto assets prior to the passage of the GENIUS Act (set $\mathcal{G}^E$), where we define “engaging” as either (i) issuing a crypto asset or (ii) offering services that use an existing crypto asset. We provide further details on our

¹⁵We also add American Express, which GICS classifies as a bank, since it operates one of the largest card networks.

payment firms and the sub-group definitions in Appendix B.

Prediction market-implied probabilities. We use data from prediction markets to estimate the extent to which markets already expected the GENIUS Act to become law, prior to the vote occurring. In prediction markets, participants trade contracts whose payoffs depend on unknown future events (Wolfers and Zitzewitz, 2004).¹⁶ These markets are useful in our context because they provide a high-frequency, numerical proxy for markets’ expectations. We use probabilities from Polymarket’s market “U.S. enacts stablecoin bill in 2025?”.

3.3 Specifications

In this section, we present the specifications that we use to test our hypotheses. In each case, we estimate coefficients γ that are the empirical counterparts of the hypothetical quantities $\bar{\eta}_t$ in Section 2.

Baseline. To test Hypothesis 1, we compare the evolution of the stock prices of payment firms and other firms in the financial sector (e.g., banks, asset managers, insurers) over the 10 trading hours surrounding the decisive House of Representatives vote on S.1582. We estimate the following difference-in-differences specification:

$$\ln(P_{i,t}) = \alpha_i + \alpha_t + \sum_{\substack{\tau=t^*-20 \\ \tau \neq t^*-1}}^{t^*+20} \gamma_\tau (\mathbf{1}_{\{t=\tau\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}}) + \epsilon_{i,t} \quad (2)$$

where $P_{i,t}$ is the stock price of firm i at time t and α_i and α_t are firm and time fixed effects. $\mathcal{G}^P$ is the set of all payment firms, and $\mathbf{1}_{\{i \in \mathcal{G}^P\}}$ is a dummy equal to one if firm i is in $\mathcal{G}^P$. We define t

¹⁶The resulting prices closely approximate the average beliefs of traders for a broad class of models (Wolfers and Zitzewitz, 2006). Data from such markets have been used to inform economic theory since at least Weitzman (1965) and such markets are widely recognized as a potent research tool (Arrow et al., 2008). A growing literature documents that they rapidly incorporate new information and exhibit lower errors than both professional forecasters and polls of the public (e.g., Rhode and Strumpf, 2004; Snowberg, Wolfers, and Zitzewitz, 2007; Berg, Forsythe, Nelson, and Rietz, 2008; Snowberg, Wolfers, and Zitzewitz, 2013; Swanson, Wang, and Wu, 2025; Diercks, Katz, and Wright, 2026).

using 15-minute intervals during regular trading hours, dropping periods when markets are closed. t^* denotes the end of the interval within which the decisive vote occurred. The resulting estimates $\gamma_\tau \times 100$ reflect the differential cumulative return, in percentage points, of payment firms relative to other financial sector firms between the period prior to the decisive vote and period τ . We cluster standard errors by firm i and by period t .

The difference-in-differences specification allows us to account for any aggregate shocks affecting the financial sector. The high-frequency approach reduces the chance that any other contemporaneous shocks occur that differentially affect payment firms relative to other financial sector firms. To estimate the average difference in returns between the pre-vote and post-vote periods, we estimate the following specification:

$$\ln(P_{i,t}) = \alpha_i + \alpha_t + \gamma \left(\mathbf{1}_{\{t \geq t^*\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}} \right) + \epsilon_{i,t} . \quad (3)$$

Heterogeneity. To test Hypotheses 2 to 4, we examine differential returns across the firm groups described in Section 3.2. We estimate the following specification:

$$\begin{aligned} \ln(P_{i,t}) = \alpha_i + \alpha_t + & \sum_{\substack{\tau=t^*-20 \\ \tau \neq t^*-1}}^{t^*+20} \gamma_\tau^{-g} \left(\mathbf{1}_{\{t=\tau\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}} \times \mathbf{1}_{\{i \notin \mathcal{G}\}} \right) \\ & + \sum_{\substack{\tau=t^*-20 \\ \tau \neq t^*-1}}^{t^*+20} \gamma_\tau^g \left(\mathbf{1}_{\{t=\tau\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}} \times \mathbf{1}_{\{i \in \mathcal{G}\}} \right) + \epsilon_{i,t} \end{aligned} \quad (4)$$

for each of $g = \mathcal{G}^X, \mathcal{G}^N, \mathcal{G}^E$. Intuitively, we repeat specification (2) but allow the estimates γ_τ^g for the subset of payment firms under consideration (e.g., firms focused on cross-border payments) to differ from the estimates γ_τ^{-g} for other payment firms.¹⁷ To test whether any such differences are statistically significant on average over our event window we estimate the following difference-in-

¹⁷When examining Hypothesis 3—i.e., when setting $g = \mathcal{G}^N$ —we drop the cross-border payments firms ($i \in \mathcal{G}^X$) from the sample to allow a clean comparison of firms focused on domestic payments that do versus do not operate networks.

differences specification:

$$\begin{aligned} \ln(P_{i,t}) = & \alpha_i + \alpha_t + \gamma_1 (\mathbf{1}_{\{t \geq t^*\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}}) \\ & + \gamma_2 (\mathbf{1}_{\{t \geq t^*\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}} \times \mathbf{1}_{\{i \in \mathcal{G}\}}) + \epsilon_{i,t} . \end{aligned} \quad (5)$$

4 Results

This section shows how equity markets repriced incumbent payment firms' stocks after the House of Representatives passed S.1582. We first present our baseline results and assess their robustness, then we document heterogeneity and benchmark our estimates against other historical shocks affecting payment firms.

4.1 Baseline

Figure 1b plots the change in average stock prices of payment firms and other financial sector firms around the passage of S.1582 by the House. Figure 1c plots our baseline estimates, γ_τ from equation (2), which compare the evolution of stock returns for payment firms against those of non-payment financial firms around the decisive House vote. Differential returns in the pre-event window are not statistically significant, supporting the interpretation that post-event differences reflect event-related repricing rather than the continuation of a prior trend.

After the vote, payment firms' stock prices fall substantially relative to those of non-payment financial firms and remain lower for the remainder of the event window, consistent with the prediction of Hypothesis 1. Across these five hours, incumbent payment firms' stock returns are on average 0.75 percentage points lower than those of other financial firms, and these differences are significant at the 1% level (Table 1, Column 1). Weighting firms by market capitalization, the average difference is 1.3 percentage points (Column 2). Multiplying this by the ex-ante market capitalization of the incumbent payment firms, these lower returns equate to approximately \$21.5 billion of lost market capitalization.

We consider the robustness of these estimates in Appendix C. We first show that they are robust to (i) using alternative definitions of the control group, (ii) controlling for differential trends by firm size and profitability, and (iii) using a narrower or wider event window. We also examine a broader set of congressional events that advanced the GENIUS Act—instead of focusing only on the final, decisive vote in the House of Representatives—and find similar results. Finally, we consider other crypto-related legislation and confirm that our estimates reflect the specific impact of the vote on the GENIUS Act.

Our baseline estimates do not reflect the full expected impact of the overall policy, i.e., of the GENIUS Act becoming law. Returning to our conceptual framework in Section 2, our baseline estimates reflect only the incremental impact of market participants updating the probability that they assign to the policy change occurring. To calculate the full expected impact of the policy, we need to scale up our estimates to correspond to a probability change from zero to one, rather than from p_{t^*-1} to p_t . The Polymarket market “U.S. enacts stablecoin bill in 2025?” implied an average probability of approximately 93% across the five hours before the vote and approximately 100% across the subsequent five hours, as shown in Figure 1a. Following equation (1), we therefore scale our estimates by $\frac{1}{1-0.93} \approx 14$.¹⁸ Applying this scaling, we estimate that in total the passage of the GENIUS Act reduced the market capitalization of incumbent payment firms by approximately 18% or \$300 billion.¹⁹

4.2 Heterogeneity

Cross-border payments. As mentioned in Section 2, the greater inefficiencies of cross-border payments, combined with the inherently borderless blockchain infrastructure underpinning stable-

¹⁸Since the prediction market only takes into account the GENIUS Act passing in the remainder of the year, our figure of 93% is actually an estimate of a quantity slightly smaller than our target p_{t^*-1} , which is the probability of the GENIUS Act passing *at any time*. Thus, our estimate of the denominator is biased upwards and so our overall estimate can be considered a lower bound.

¹⁹Our scaling methodology, following Snowberg, Wolfers, and Zitzewitz (2011), assumes that each incremental change in probability has the same impact on stock prices. While it is conceivable that the relationship could instead be non-linear, we find that our estimated elasticities remain similar when estimating the average differential response using additional events that took place at different probability levels (Figure C.1c) and when using alternative windows containing probability changes of different sizes (Table C.1).

coins, make cross-border payment firms especially exposed to competition from stablecoin-based challengers. Indeed, Figure 2a and column 3 in Table 1 show that cross-border payments firms saw significantly larger stock price declines than other payment firms. After the vote, stock prices of non-cross-border payment firms declined by approximately 0.67 percent relative to non-payment financial firms. Cross-border payment firms saw an additional decline of 0.64 percentage points, for a total decline of approximately 1.3 percent.²⁰

When weighting the regression results by market capitalization, the total decline for cross-border payment firms is approximately 1.9 percent. Scaling this 1.9 percent to account for anticipation, as described in Section 4.1, we estimate that Congress passing the GENIUS Act reduced the total stock market value of cross-border payment firms by approximately 27 percent. This substantial impact supports Hypothesis 2 and suggests that stablecoins could play a major role in cross-border payments in future.

Network operators. Figure 2b shows that, among firms focused on domestic payments, those that operate networks did not see a statistically significant decline in market value relative to other financial sector firms. Column 4 in Table 1 shows that the returns of such firms after the vote were significantly above the average for incumbent payment firms in general. These results support Hypothesis 3, suggesting that network effects may help insulate payment firms from competition by stablecoin-based challengers.

Early engagement with crypto assets. Lastly, Figure 2c shows that incumbent payment firms that engaged early with blockchain technology and offered crypto-related services did not see a significant decline in market value relative to other financial sector firms. Column 5 in Table 1 shows that the returns of such firms after the vote were significantly above the average for incumbent payment firms in general. These results support Hypothesis 4 and suggest that crypto-engaged payment firms may be better positioned to seize emerging opportunities or to respond effectively

²⁰While our sample contains only a small number of specialized cross-border payment firms, this result is not driven by just one firm. Figure A.2 plots the changes in value of all our cross-border payment firms and reveals that they all see substantial price declines after the vote.

to new competition from stablecoin-based challengers.

4.3 Benchmarking our findings

Finally, we benchmark our findings against other historical shocks to payment firms. Figure 3 summarizes the results and Appendix D explains the construction of the benchmarks. In short, we compare our findings to: (a) the impact on U.S. payment firms of the Durbin Amendment, an unexpected last-minute amendment to the Dodd-Frank Act in 2010 that capped debit card interchange fees; (b) the impact on U.S. payment networks of President Trump announcing his support for the Credit Card Competition Act in January 2026; (c) the impact on U.S. and European payment firms between 2016 and 2022 of the anticipated launch of the digital euro; (d) the impact on One97 Communications (the parent company of Indian fintech firm Paytm) of the Reserve Bank of India directing Paytm Payments Bank to stop accepting fresh deposits in January 2024; and (e) the impact on a major global mobile payments platform of its IPO being suspended and more stringent regulation being imposed upon it.

We find that the impact of U.S. stablecoin legislation on U.S. payment firms was larger than the impacts on U.S. payment firms of recent pro-competitive regulatory shocks (i.e., (a), (b) and (c) above), but smaller than the impacts of severe regulatory actions that targeted specific non-U.S. payment firms (i.e., (d) and (e)).

Given this relatively large estimated impact, one might naturally expect incumbent payment firms to respond by engaging more closely with stablecoins (see, for example, Goldstein, 2023). In Appendix E, we provide descriptive evidence that this is indeed the case. The months immediately after the GENIUS Act passed saw increases in (i) the share of payment firms offering crypto-based services and (ii) the frequency with which stablecoins are mentioned on payment firms' earnings calls.

5 Conclusion

We contribute novel, forward-looking empirical evidence to a longstanding debate on the role of crypto assets in payments. Using high-frequency stock price data, we find that financial market participants anticipate that stablecoins will play a significant role in the payments ecosystem. Following the decisive congressional vote on the GENIUS Act—which supported the use of stablecoins in payments—the market value of incumbent payments firms declined significantly, signaling that the new framework is expected to heighten competition in payments.

Combining the stock price data with information from prediction markets, we estimate that U.S. legislation supporting the use of stablecoins in payments reduced the market value of listed incumbent payment firms by 18% or approximately \$300 billion. Benchmarking these results against estimates for other historical shocks, we find that they are slightly larger than the impacts on U.S. payment firms of recent pro-competitive regulatory shocks such as the Durbin Amendment and the European Central Bank’s plan to issue a digital euro.

We find substantial variation in the estimated impact across different types of payment firms: (i) it is larger for firms specializing in cross-border payments, consistent with stablecoins’ comparative advantage in this context; (ii) it is smaller for incumbents protected by network effects, consistent with these insulating them from competitive pressures; and (iii) it is smaller for incumbents already offering crypto-related services, consistent with early engagement with the technology improving firms’ ability to exploit opportunities that arise or to respond to new competition.

References

- A16Z CRYPTO (2025): “State of Crypto 2025,” Tech. rep., Andreessen Horowitz.
- ACEMOGLU, D., S. JOHNSON, A. KERMANI, J. KWAK, AND T. MITTON (2016): “The value of connections in turbulent times: Evidence from the United States,” Journal of Financial Economics, 121, 368–391.
- ADRIAN, P. T., P. BAINS, M. BECHARA, E. CERUTTI, S. FORTE, F. GRINBERG, A. GULLO, M. HENGGE, A. JEKABSONE, K. KAO, T. M. GRIFFOLI, S. M. PERIA, M. MICCOLI, M. REUTER, AND N. SUGIMOTO (2025): “Understanding Stablecoins,” IMF Departmental Papers.
- AGUR, I., G. VILLEGAS-BAUER, T. MANCINI-GRIFFOLI, M. S. MARTINEZ PERIA, AND B. TAN (2025): Tokenization and financial market inefficiencies, International Monetary Fund.
- AHMED, R. AND I. ALDASORO (2025): “Stablecoins and safe asset prices,” Tech. rep., Bank for International Settlements.
- AIELLO, D., S. R. BAKER, T. BALYUK, M. D. MAGGIO, M. J. JOHNSON, AND J. D. KOTTER (2023a): “The Effects of Cryptocurrency Wealth on Household Consumption and Investment,” NBER Working Papers 31445, National Bureau of Economic Research.
- (2023b): “Who Invests in Crypto? Wealth, Financial Constraints, and Risk Attitudes,” NBER Working Papers 31856, National Bureau of Economic Research.
- ALLAIRE, J. (2025): “The Internet’s Next Currency,” Interview by Steven Levy. *Wired*, December 4, 2025. Video, 12:45.
- ALLIUM LABS (2025): “Onchain Analytics Dashboard,” Available at: <https://docs.allium.so/historical-data/stablecoinsadjusted-metrics>.

- ALVAREZ, F. E., D. ARGENTE, F. LIPPI, E. MÉNDEZ, AND D. VAN PATTEN (2023): “Strategic Complementarities in a Dynamic Model of Technology Adoption: P2P Digital Payments,” NBER Working Papers 31280, National Bureau of Economic Research, Inc.
- ARROW, K. J., R. FORSYTHE, M. GORHAM, R. HAHN, R. HANSON, J. O. LEDYARD, S. LEV-MORE, R. LITAN, P. MILGROM, F. D. NELSON, G. R. NEUMANN, M. OTTAVIANI, T. C. SCHELLING, R. J. SHILLER, V. L. SMITH, E. SNOWBERG, C. R. SUNSTEIN, P. C. TETLOCK, P. E. TETLOCK, H. R. VARIAN, J. WOLFERS, AND E. ZITZEWITZ (2008): “The Promise of Prediction Markets,” Science, 320, 877–878, American Association for the Advancement of Science.
- AUER, R., U. LEWRICK, AND J. PAULICK (2025): “DeFying gravity? An empirical analysis of cross-border Bitcoin, Ether and stablecoin flows,” BIS Working Papers 1265, Bank for International Settlements, Basel, Switzerland.
- BACK, A. (2002): “Hashcash - A Denial of Service Counter-Measure,” .
- BAUER, M. D. AND E. T. SWANSON (2023): “An Alternative Explanation for the “Fed Information Effect”,” American Economic Review, 113, 664–700.
- BERG, J., R. FORSYTHE, F. NELSON, AND T. RIETZ (2008): “Results from a Dozen Years of Election Futures Markets Research,” Handbook of Experimental Economics Results, 1, Part 5, 742–751, Elsevier.
- BERG, T., J. KEIL, F. MARTINI, AND M. PURI (2024): “CBDCs, Payment Firms, and Geopolitics,” NBER Working Papers 32857, National Bureau of Economic Research, Inc.
- BERTSCH, C. (2023): “Stablecoins: Adoption and Fragility,” SSRN Electronic Journal.
- BUDISH, E. (2025): “Trust at Scale: The Economic Limits of Cryptocurrencies and Blockchains,” The Quarterly Journal of Economics, 140, 1–62, President and Fellows of Harvard College.

- BURLON, L., M. A. MUÑOZ, AND F. SMETS (2024): “The Optimal Quantity of CBDC in a Bank-Based Economy,” American Economic Journal: Macroeconomics, 16, 172–217.
- CASINI, A. AND A. MCCLOSKEY (2025): “Identification and Estimation of Causal Effects in High-Frequency Event Studies,” ArXiv:2406.15667 [econ].
- CERUTTI, E., J. CHEN, AND M. HENGGE (2025): “A primer on bitcoin cross-border flows: Measurement and drivers,” Journal of International Money and Finance, 159, 103424.
- CERUTTI, E., M. FIRAT, M. HENGGE, AND T. SAGAWA (2026): “Stablecoin Shocks,” IMF Working Papers.
- CHAUM, D. (1983): “Blind Signatures for Untraceable Payments,” in Advances in Cryptology, ed. by D. Chaum, R. L. Rivest, and A. T. Sherman, Boston, MA: Springer US, 199–203.
- CHIU, J. AND T. V. KOEPPL (2019): “Blockchain-Based Settlement for Asset Trading,” The Review of Financial Studies, 32, 1716–1753.
- CHO, T., M. GROTTIERA, L. KREMENS, AND H. KUNG (2026): “The Present Value of Future Market Power,” The Review of Financial Studies.
- CONG, L. W. AND Z. HE (2019): “Blockchain Disruption and Smart Contracts,” The Review of Financial Studies, 32, 1754–1797.
- COPESTAKE, A., D. KIRTI, AND M. MARTINEZ PERIA (2025a): “Growing Retail Digital Payments: The Value of Interoperability,” IMF Fintech Notes, 2025, International Monetary Fund.
- COPESTAKE, A., D. KIRTI, M. MARTINEZ PERIA, AND Y. ZENG (2025b): “Integrating Fragmented Networks: The Value of Interoperability in Money and Payments,” IMF Working Papers, 2025, 1, International Monetary Fund.
- CROUZET, N., A. GUPTA, AND F. MEZZANOTTI (2023): “Shocks and Technology Adoption: Evidence from Electronic Payment Systems,” Journal of Political Economy, 131.

DAI, W. (1998): “b-money,” .

D’AVERNAS, A., V. MAURIN, AND Q. VANDEWEYER (2022): “Can Stablecoins Be Stable?” SSRN Electronic Journal.

DIERCKS, A. M., J. D. KATZ, AND J. H. WRIGHT (2026): “Kalshi and the Rise of Macro Markets,” NBER Working Papers 34702, National Bureau of Economic Research.

DRECHSEL, T. AND K. MIURA (2025): “The macroeconomic effects of bank regulation: New evidence from a high-frequency approach,” .

EBRAHIMI, Z., M. GUENNEWIG, B. ROUTLEDGE, AND A. ZETLIN-JONES (2025): “Achieving Consensus on Blockchains,” CRC TR 224 Discussion Paper Series 224/2025/685, University of Bonn and University of Mannheim, Germany.

FERRARI MINESSO, M. AND D. SIENA (2026): “Private money and public debt. U.S. Stablecoins and the global safe asset channel,” Working Paper Series, Number: 2787 European Central Bank.

FINNEY, H. (2004): “Reusable proofs of work (RPOW),” .

FRINGUELLOTTI, F. AND T. KROEN (2024): “Payout Restrictions and Bank Risk-Shifting,” SSRN Scholarly Paper 4969786, Social Science Research Network, Rochester, NY.

FSB (2025): “G20 Roadmap for Enhancing Cross-border payments: Consolidated progress report for 2025,” Priority actions for achieving the G20 targets.

GOLDSTEIN, I. (2023): “Information in Financial Markets and Its Real Effects,” Review of Finance, 27, 1–32.

GORTON, G. B., E. C. KLEE, C. P. ROSS, S. Y. ROSS, AND A. P. VARDOULAKIS (2025): “Leverage and Stablecoin Pegs,” Journal of Financial and Quantitative Analysis, 1–38.

- GORTON, G. B., C. P. ROSS, AND S. Y. ROSS (2022): “Making Money,” NBER Working Papers 29710, National Bureau of Economic Research.
- GRAF VON LUCKNER, C., R. KOEPKE, AND S. SGHERRI (2024): “Crypto as a Marketplace for Capital Flight,” Tech. Rep. WP/24/133, International Monetary Fund, Washington, D.C.
- GRAF VON LUCKNER, C., C. M. REINHART, AND K. ROGOFF (2023): “Decrypting new age international capital flows,” Journal of Monetary Economics, 138, 104–122.
- GÜRKAYNAK, R. S., B. SACK, AND E. SWANSON (2005): “The Sensitivity of Long-Term Interest Rates to Economic News: Evidence and Implications for Macroeconomic Models,” American Economic Review, 95, 425–436.
- HAZELL, J. AND S. HOBLER (2024): “Do Deficits Cause Inflation? A High Frequency Narrative Approach,” Discussion Papers 2439, Centre for Macroeconomics (CFM).
- HE, H., Y. ZHAO, AND D. ZHOU (2026): “Is Stablecoin Really Stable? A DSGE Investigation,” IMF Working Papers.
- HIGGINS, S. (2024): “Financial Technology Adoption: Network Externalities of Cashless Payments in Mexico,” American Economic Review, 114, 3469–3512.
- HINZEN, F. J., K. JOHN, AND F. SALEH (2022): “Bitcoin’s Limited Adoption Problem,” SSRN Scholarly Paper 3334262, Social Science Research Network, Rochester, NY.
- JAROCIŃSKI, M. AND P. KARADI (2020): “Deconstructing Monetary Policy Surprises— The Role of Information Shocks,” American Economic Journal: Macroeconomics, 12, 1–43.
- KAY, B. S., M. D. MANUSZAK, AND C. M. VOJTECH (2018): “Competition and complementarities in retail banking: Evidence from debit card interchange regulation,” Journal of Financial Intermediation, 34, 91–108.
- KOGAN, L., D. PAPANIKOLAOU, A. SERU, AND N. STOFFMAN (2017): “Technological Innovation, Resource Allocation, and Growth,” The Quarterly Journal of Economics, 132, 665–712.

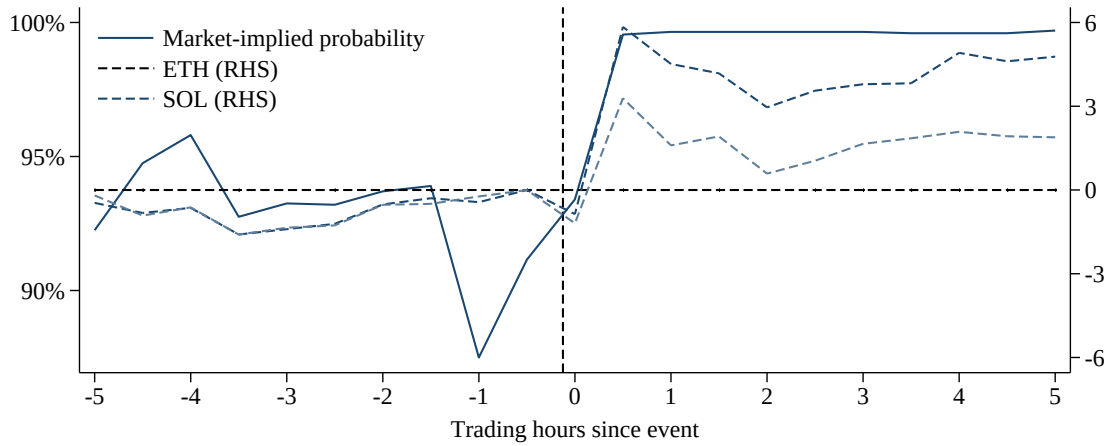
- KRUGMAN, P. (2025): “Crypto Is Still for Criming,” Substack newsletter post.
- KUTTNER, K. N. (2001): “Monetary policy surprises and interest rates: Evidence from the Fed funds futures market,” Journal of Monetary Economics, 47, 523–544.
- KÄNZIG, D. R. (2021): “The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements,” American Economic Review, 111, 1092–1125.
- LI, J. AND W. MANN (2025): “Digital Tokens and Platform Building,” The Review of Financial Studies, 38, 1921–1954.
- LI, Y. AND S. MAYER (2025): “Financial Intermediation with Fragile Seigniorage: A Dynamic Model of Stablecoin Issuers,” .
- LIAO, G., T. HADEED, AND Z. ZENG (2023): “Beyond Speculation: Payment Stablecoins for Real-time Gross Settlements,” SSRN Scholarly Paper 4476859, Social Science Research Network, Rochester, NY.
- MA, Y., Y. ZENG, AND A. L. ZHANG (2025): “Stablecoin Runs and the Centralization of Arbitrage,” SSRN Scholarly Paper 5285654, Social Science Research Network, Rochester, NY.
- MUKHARLYAMOV, V. AND N. SARIN (2025): “Price regulation in two-sided markets: Empirical evidence from debit cards,” Journal of Financial Economics, 172, 104094.
- NAKAMOTO, S. (2008): “Bitcoin: A Peer-to-Peer Electronic Cash System,” 9.
- NAKAMURA, E. AND J. STEINSSON (2018): “High-Frequency Identification of Monetary Non-Neutrality: The Information Effect*,” The Quarterly Journal of Economics, 133, 1283–1330.
- ONE 97 COMMUNICATIONS LIMITED (2024): “Disclosure under Regulation 30 of the SEBI (Listing Obligations and Disclosure Requirements) Regulations, 2015 – Press Release,” .
- RESERVE BANK OF INDIA (2024): “Action against Paytm Payments Bank Ltd under Section 35A of the Banking Regulation Act, 1949,” Published: Press Release 2023–2024/1774.

- REUTER, M. (2024): “Platform Precommitment via Decentralization,” IMF Working Papers, 2024, 1.
- RHODE, P. W. AND K. S. STRUMPF (2004): “Historical Presidential Betting Markets,” Journal of Economic Perspectives, 18, 127–141.
- ROCHET, J. AND J. TIROLE (2004): “Two-Sided Markets: An Overview,” Toulouse, France.
- ROCHET, J.-C. AND J. TIROLE (2003): “Platform Competition in Two-Sided Markets,” Journal of the European Economic Association, 1, 990–1029, MIT Press.
- ROGOFF, K. (2025a): “The Past and Future of Money: New Technologies and Economic Risks,” Group of Thirty Report, iISBN: 1-56708-192-4.
- ROGOFF, K. S. (2025b): Our dollar, your problem: an insider’s view of seven turbulent decades of global finance, and the road ahead, The Henry L. Stimson lectures series, New Haven: Yale University Press.
- SACKS, D. (2025): “AI, Crypto, China, Dems, and SF,” The a16z Show, available at: <https://open.spotify.com/episode/5OndRhYfJK9BB5aYKe5IUv>.
- SALEH, F. (2021): “Blockchain without Waste: Proof-of-Stake,” The Review of Financial Studies, 34, 1156–1190.
- SMETS, F. (2025): “Cross-border payments—a catalyst for global integration and growth,” SEACEN 61st Governors Conference.
- SNOWBERG, E., J. WOLFERS, AND E. ZITZEWITZ (2007): “Partisan Impacts on the Economy: Evidence from Prediction Markets and Close Elections,” The Quarterly Journal of Economics, 122, 807–829, President and Fellows of Harvard College.
- (2011): “How Prediction Markets Can Save Event Studies,” Working Paper 16949, National Bureau of Economic Research.

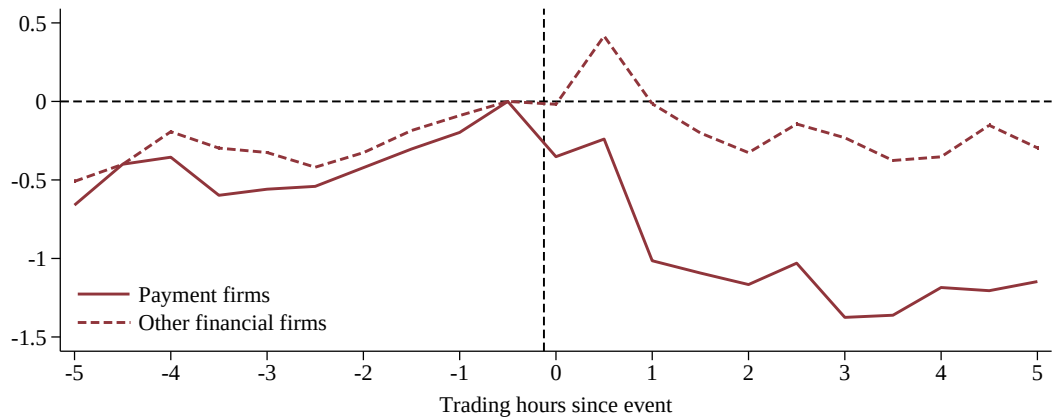
- (2013): “Prediction Markets for Economic Forecasting,” in Handbook of Economic Forecasting, ed. by G. Elliott and A. Timmermann, Elsevier, vol. 2 of Handbook of Economic Forecasting, 657–687, iSSN: 1574-0706.
- SWANSON, E. T. (2021): “Measuring the effects of federal reserve forward guidance and asset purchases on financial markets,” Journal of Monetary Economics, 118, 32–53.
- SWANSON, E. T., R. WANG, AND Y. WU (2025): “The Effects of Monetary Policy on Macroeconomic Expectations: High-Frequency Evidence from Traded Event Contracts,” .
- SZABO, N. (2005): “Bit gold,” .
- THE WHITE HOUSE (2025): “Fact Sheet: President Donald J. Trump Signs GENIUS Act into Law,” .
- WANG, L. (2025): “Regulating Competing Payment Networks,” SSRN Scholarly Paper 5262161, Social Science Research Network, Rochester, NY.
- WEBER, M., B. CANDIA, O. COIBION, AND Y. GORODNICHENKO (2023): “Do You Even Crypto, Bro? Cryptocurrencies in Household Finance,” NBER Working Papers 31284, National Bureau of Economic Research.
- WEINBERG, J. A. (1997): “The Organization of Private Payment Networks,” SSRN Scholarly Paper 2129857, Social Science Research Network, Rochester, NY.
- WEITZMAN, M. (1965): “Utility Analysis and Group Behavior: An Empirical Study,” Journal of Political Economy, 73, 18–26, University of Chicago Press.
- WOLFERS, J. AND E. ZITSEWITZ (2004): “Prediction Markets,” Journal of Economic Perspectives, 18, 107–126.
- (2006): “Interpreting Prediction Market Prices as Probabilities,” Working Paper 12200, National Bureau of Economic Research.

Figure 1: Evolution of market values around the House passage of S.1582

(a) Prediction market-implied probability and changes in crypto asset prices (percent)



(b) Average price response by group of firms (percent)

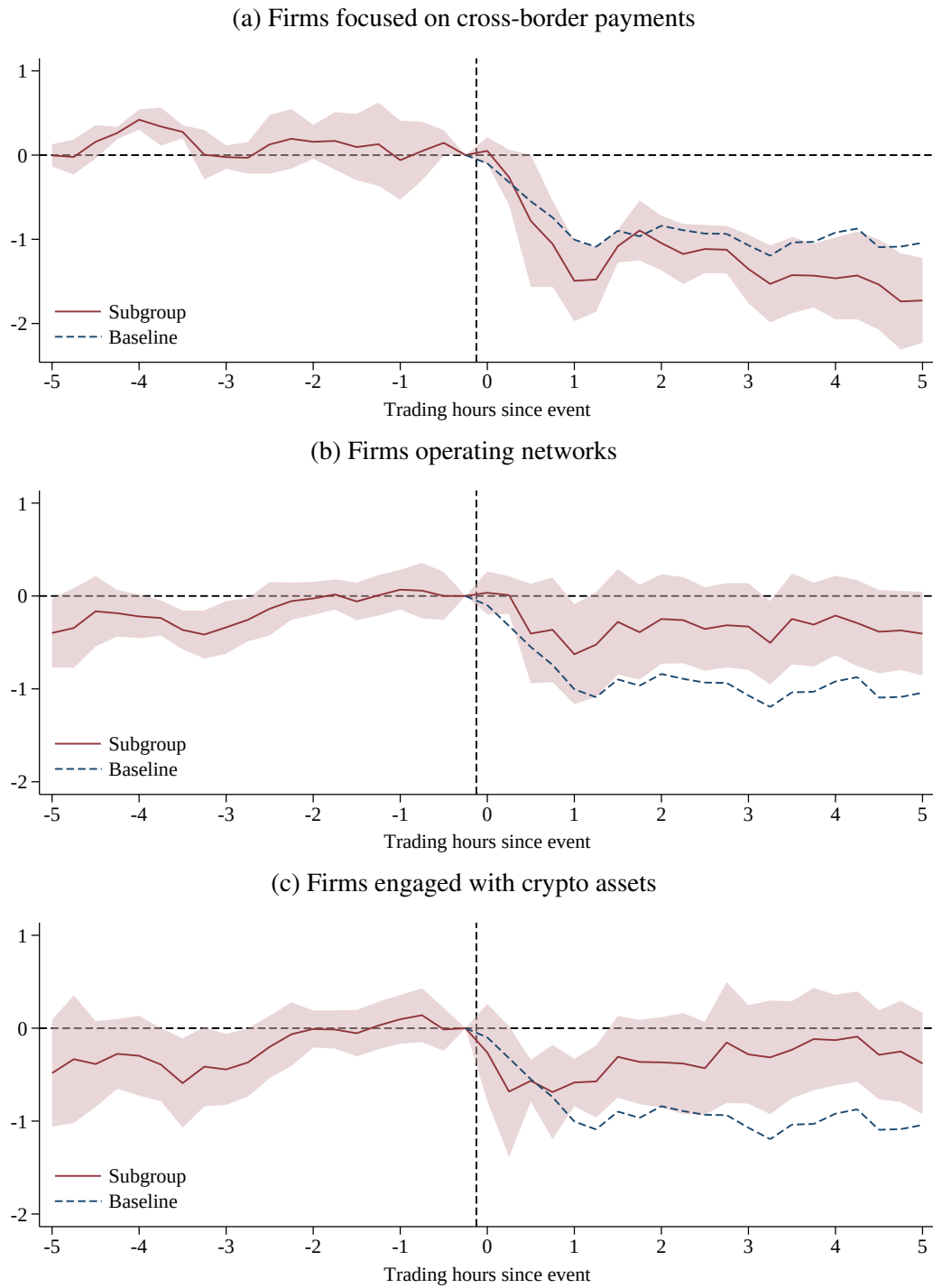


(c) Differential return of payment firms relative to other financial firms (percentage points)



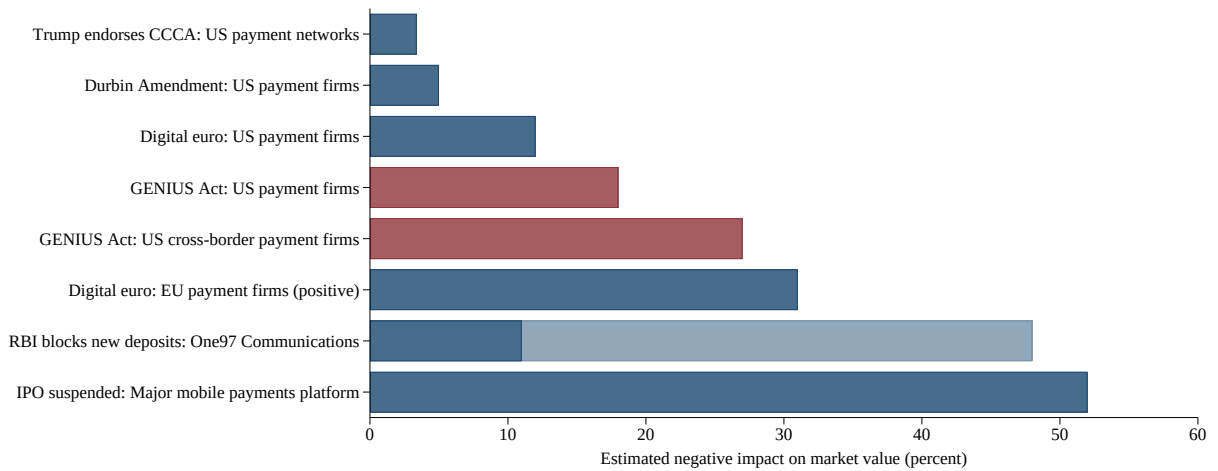
Notes: Panel (a) shows the probability of stablecoin legislation passing implied by the Polymarket market “U.S. enacts stablecoin bill in 2025?”, along with the percent change in price of the native assets of the Ethereum and Solana blockchains relative to the period directly prior to the vote. Panel (b) shows the change in the average prices of payment firms and other financial firms after the vote, in percent. Panel (c) shows our baseline results, estimated using equation (2). While both crypto and prediction markets operate 24/7, we show all prices for the same set of stock-market trading hours so that all series are directly comparable.

Figure 2: Differential return around House passage of S.1582, by firm group (percentage points)



Notes: Each panel plots event-time coefficients from equation (4) estimated at 15-minute frequency around the decisive House vote advancing S.1582. Panel (a) shows the differential response of firms whose primary business relates to cross-border payments. Panel (b) shows the differential response of firms whose primary business involves operating one or more networks. Panel (c) shows the differential response of firms that were early to engage with crypto assets, as defined in Appendix B. In each case, the blue dashed line plots our baseline coefficients from Figure 1c.

Figure 3: Comparing our results to other historical shocks to payment firms



Notes: This figure benchmarks the negative impacts on the market value of US payment firms that we find against the estimated impacts of other historical shocks affecting payment firms. The red bars show our estimates of the impact of U.S. stablecoin legislation, and the blue bars show the estimated impacts of other historical shocks. Specifically, the blue bars reflect, in turn: (i) the impact on U.S. payment network operators of President Trump announcing his support for the Credit Card Competition Act in January 2026; (ii) the impact on U.S. payment firms of the Durbin Amendment, an unexpected last-minute amendment to the Dodd-Frank Act in 2010 that capped debit card interchange fees; (iii) the impact on U.S. payment firms between 2016 and 2022 of the anticipated launch of the digital euro; (iv) the impact on European payment firms of the anticipated launch of the digital euro, over the same period; (v) the impact on One97 Communications (the parent company of Indian fintech firm Paytm) of the Reserve Bank of India directing Paytm Payments Bank to stop accepting fresh deposits in January 2024; and (vi) the impact on a major global mobile payments platform of its IPO being suspended and more stringent regulation being imposed upon it. The estimated impact in case (iv) is positive, unlike the other cases. In case (v), the end of the dark (light) blue bar indicates the lower (upper) bound for the estimated impact. We describe the sources and construction of the benchmarks in detail in Appendix D. The estimated impacts of the Durbin Amendment and a digital euro are from [Berg, Keil, Martini, and Puri \(2024\)](#).

Table 1: Average differential return of payment firms relative to other financial firms after the House passage of S.1582 (percentage points)

	(1)	(2)	(3)	(4)	(5)
All payment firms:	-0.747***	-1.272***	-0.672**	-1.055***	-0.996***
$\mathbf{1}_{\{t \geq t^{Pass}\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}}$	(-3.17)	(-3.73)	(-2.60)	(-2.80)	(-3.34)
Cross-border:			-0.642**		
$\mathbf{1}_{\{t \geq t^{Pass}\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}} \times \mathbf{1}_{\{i \in \mathcal{G}^X\}}$			(-2.22)		
Network operators:				0.885**	
$\mathbf{1}_{\{t \geq t^{Pass}\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}} \times \mathbf{1}_{\{i \in \mathcal{G}^N\}}$				(2.03)	
Crypto-engaged:					0.845**
$\mathbf{1}_{\{t \geq t^{Pass}\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}} \times \mathbf{1}_{\{i \in \mathcal{G}^E\}}$					(2.21)
Firm FEs	✓	✓	✓	✓	✓
Period FEs	✓	✓	✓	✓	✓
Mkt. Cap Weighting		✓			
N	27,839	27,716	27,839	27,675	27,839

Notes: t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered by firm and by period. $\mathcal{G}^P$ denotes the set of all U.S.-listed payment firms, $\mathcal{G}^X$ denotes firms whose primary business relates to cross-border payments, $\mathcal{G}^N$ denotes the set of all firms whose primary business involves operating one or more networks, and $\mathcal{G}^E$ denotes firms that were early to engage with crypto assets, as defined in Appendix B. Coefficients in columns (1) and (2) reflect the average differential return of payments firms relative to other non-payments financial firms, in percent, over the five hours after the decisive House vote on S.1582. Column (1) weights all observations equally, while column (2) weights them by pre-event market capitalization. The lower coefficients in columns (3), (4), and (5) reflect the incremental difference in such returns for firms in the named category (e.g., firms focused on cross-border payments). In Column (4), we exclude cross-border payment firms from the sample as described in Section 3.3 to allow a clean comparison of firms focused on domestic payments that do versus do not operate networks.

ONLINE APPENDIX TO

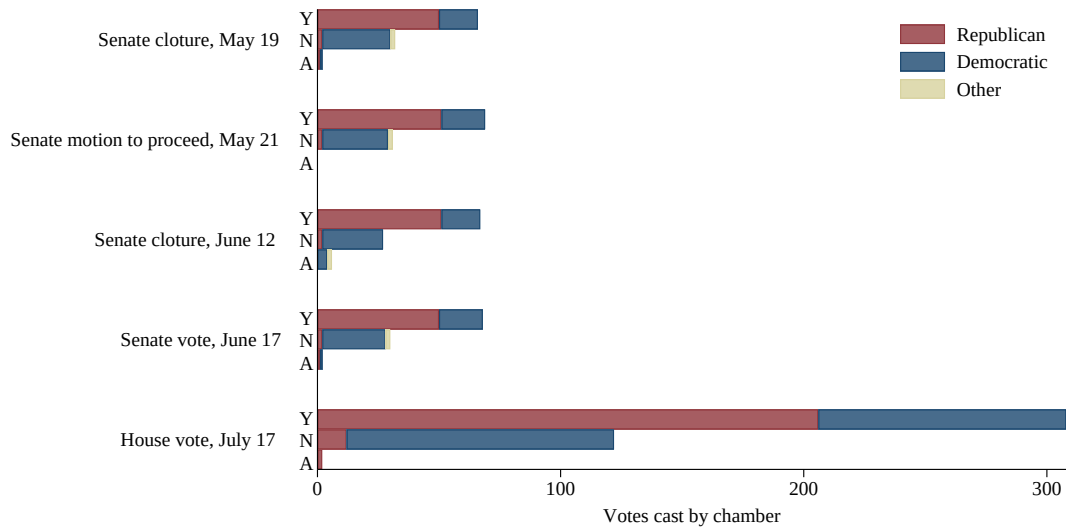
Stablecoins and the Future of Payments: Evidence from Financial Markets

by Alexander Copestake, Cage Englander,
Maria Soledad Martinez Peria, and Germán Villegas-Bauer

A	Additional figures	i
B	Payment firm groups	iii
C	Robustness	ix
D	Benchmarks	xviii
E	Additional descriptive evidence	xxii

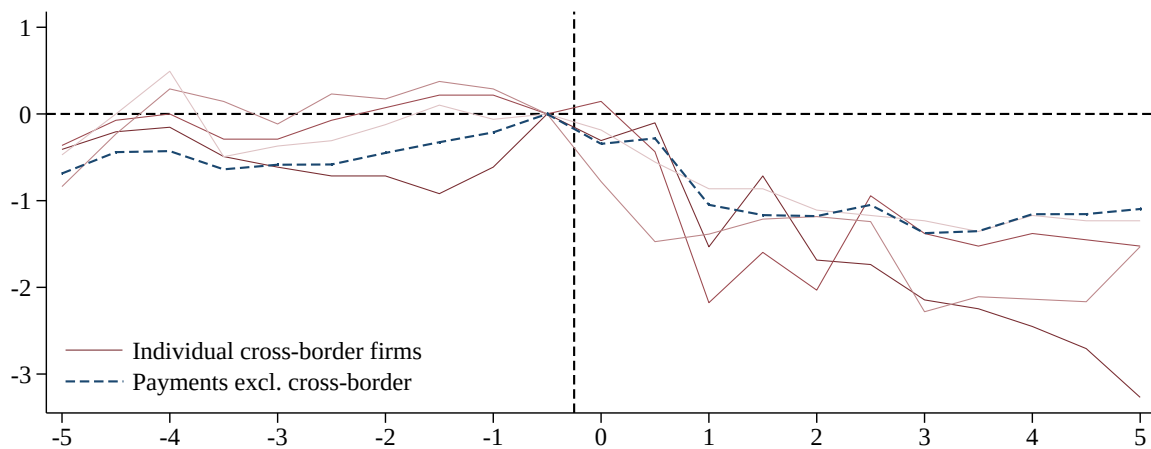
A Additional figures

Figure A.1: Votes by party on S.1582



Notes: This figure shows the vote breakdown by party for the five successful votes on the main text of S.1582, the bill that would become the GENIUS Act, during the process of its passage through Congress. “Y” denotes votes for the motion, “N” denotes votes against the motion, and “A” indicates abstentions. The larger number of votes in the House reflects the larger number of representatives in the chamber. *Source:* [Congress.gov](https://www.congress.gov).

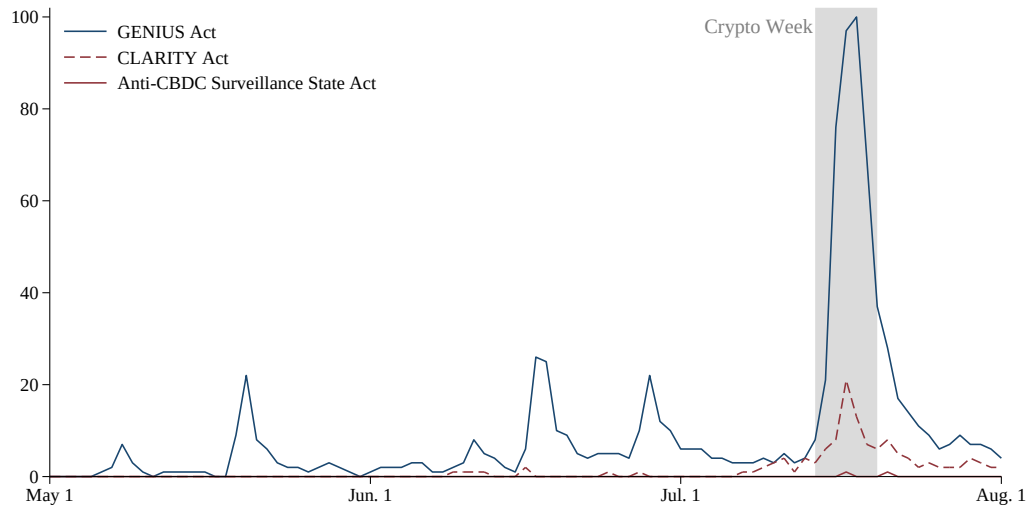
Figure A.2: Stock price changes for cross-border payment firms around House passage of S.1582



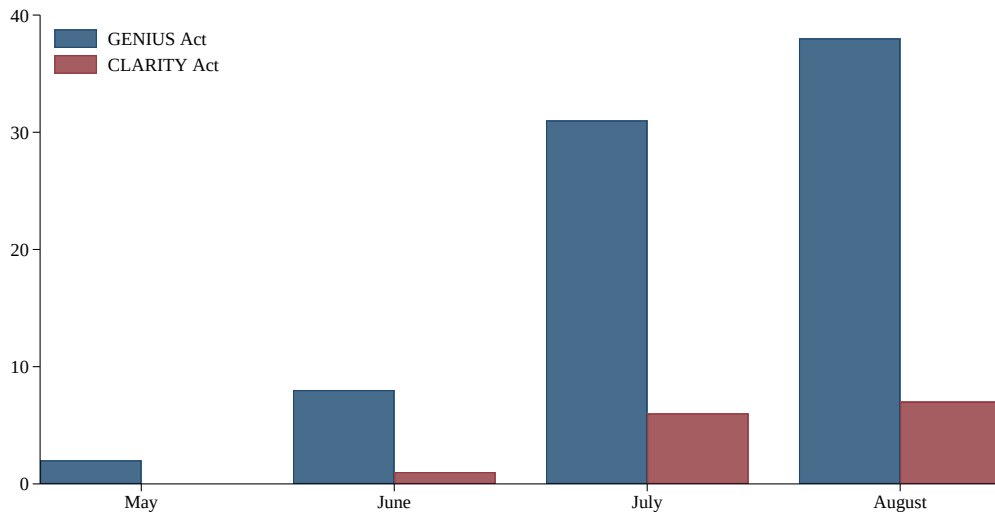
Notes: The red lines show the percentage change in the stock prices of each of the payment firms in our sample that specialize in cross-border payments, relative to the period directly prior to S.1582 passing the House of Representatives. The dashed blue line shows the change in the average stock price of all the other payments firms in our sample.

Figure A.3: Attention regarding potential crypto asset legislation

(a) Google Trends



(b) Mentions in firm communication



Notes: Panel (a) shows online search interest in the U.S. for each of the three pieces of legislation considered during “Crypto Week”. The peak search interest over the period is indexed to 100, and all other values represent search interest relative to that peak; for instance, a value of 50 means that the term is half as popular as the peak value. Panel (b) shows the number of earnings and conference calls by financial firms that mention a given piece of legislation. Specifically, we identify instances in which the phrase “GENIUS Act” or “CLARITY Act” occur. The Anti-CBDC Surveillance State Act is omitted from Panel (b) since it does not appear in any calls. *Source:* [Google Trends](#), LSEG.

B Payment firm groups

We define payment firms as those in GICS code *40201060*, which includes “providers of transaction & payment processing services and related payment services, including digital/mobile payment processors, payment service providers & gateways, and digital wallet providers.” We also classify American Express as a payment firm, rather than a bank (as it appears in GICS), since it operates one of the largest card networks globally. Table B.1 shows summary statistics on payment firms and their sub-groups.

Since no further sub-classification of payment firms exists in GICS, we manually identify the functions that each firm performs in the payments value chain (e.g., operating a bank correspondence network, providing backend processing services).²¹ The results are shown in Table B.2. We then repeat the process to identify the primary function—the one that drives the business model and/or underpins the revenue—among the functions that each firm performs. Lastly, we classify each firm into one of three categories based on these primary functions: cross-border payments specialists, domestic networks (including bank and consumer networks), and other firms (which are mostly processors or other service providers). Table B.3 shows the resulting classification.

The cross-border payments specialists in Table B.3 constitute firms whose primary business function is in column (ix) of table B.2, and we label them $\mathcal{G}^X$. Specifically, these are firms whose primary business is transferring funds from an end party in one country to an end party in a second country. Typically, they accept funds on the sending side via cash, card, or bank transfer. They then net and settle cross-border obligations through their own accounts and intermediaries such as correspondent banks, local clearing banks, and licensed payout partners (e.g., money-transfer agents), handling currency conversion when required. Funds are disbursed locally via bank deposit, wallet credit, or cash pickup, with revenue coming mainly from explicit fees and foreign exchange spreads.

²¹Specifically, for each payment firm and each core function in the payments value chain, we used a search-enabled LLM to identify whether the firm performed that specific function, and to provide supporting reasoning and sources. Before reviewing the AI outputs, we also manually examined firms’ regulatory filings and classified them ourselves. Whenever our classification differed from the AI’s, we performed additional review to reconcile the discrepancy.

Network providers (including bank and consumer networks) in Table B.3 are firms whose primary business function is in columns (i), (ii), (vii), or (viii) of Table B.2, and we label them $\mathcal{G}^N$. These firms have a primary business function that involves operating some form of network, such as a card network or an ATM network. The form of network effect from which these firms benefit varies. On one hand, peer-to-peer products (such as PayPal’s Venmo and Block’s Cash App) benefit from direct network effects, in which the value of the service to a given user grows simply with the addition of more users of the same type. On the other hand, card networks (e.g., Visa) and point-of-sale networks (e.g., Block) benefit primarily from indirect network effects. Here, the value of the service to one type of user (e.g., a customer) grows with the number of users of another type (e.g., merchants) that also participate in the service, and vice versa—such that more users of the first type attract more users of the second type, who in turn attracts more users of the first type.

Finally, we separately identify firms that had engaged with crypto assets prior to the decisive House vote on S.1582 (our set $\mathcal{G}^E$). We include a firm in this set if by the day of the vote it had either (i) issued a crypto asset or (ii) offered services utilizing an existing crypto asset. Table B.4 lists the firms engaging with crypto assets prior to the passage of the GENIUS Act and summarizes the form of engagement.

Table B.1: Descriptive statistics on payment firms

	$\mathcal{G}^P$			$\mathcal{G}^X$	$\mathcal{G}^N$	$\mathcal{G}^E$
	$p(25)$	$p(50)$	$p(75)$	$p(50)$	$p(50)$	$p(50)$
Size:						
Market capitalization (\$ million)	430	2,649	20,068	2,268	22,026	9,094
Revenue (\$ million, quarterly)	76	412	1,104	336	1,550	1,035
Total assets (\$ million)	382	2,316	14,674	4,575	6,555	6,402
Total liabilities (\$ million)	152	949	13,695	3,750	5,462	5,217
Employees (#)	1,019	2,800	11,372	2,603	15,686	9,306
Profitability:						
Net income (\$ million, quarterly)	-0.37	14.71	122.10	15.24	97.6	86.3
Return on assets	0.33	2.65	9.10	5.21	5.71	6.83
Return on common equity	-4.64	14.39	32.39	25	22.94	27.78
Valuation:						
Price/earnings ratio	14.58	26.93	65.75	9.95	25.77	33.1

Notes: This table shows balance sheet and income data for incumbent payment firms, as defined in Section 3.2. Values are based on the most recent observation prior to the House of Representatives vote on July 17, 2025. The first three columns show the 25th, 50th, and 75th percentiles of each variable for payment firms broadly. The following three columns show the 50th percentile of each variable for cross-border, network, and crypto-engaged firms, respectively.

Table B.2: Payment firms and their functions

	Closed-loop operator	Open-loop operator	Open-loop participant	Processor	Inter-party services	Intra-party services	Remote gateway	Physical POS	Cross- border
Affirm Holdings Inc	0	0	1	0	1	0	0	0	0
American Express Inc	0	0	1	0	1	1	0	0	1
Block Inc	1	0	1	1	1	1	1	1	1
Cantaloupe Inc	0	0	0	1	0	0	0	1	0
Cass Information Systems Inc	1	0	1	1	0	1	0	0	1
Change Financial Ltd	0	0	1	1	1	0	0	1	0
Chime Financial Inc	0	0	1	1	0	0	0	0	0
Corpay Inc	1	0	1	1	1	1	0	0	1
Euronet Worldwide Inc	1	0	1	1	0	1	0	1	1
FIS Inc	1	0	1	1	1	1	0	0	1
Flywire Corp	0	0	0	1	1	0	1	0	1
Global Payments Inc	1	0	0	1	1	0	1	1	1
International Money Express Inc	0	0	1	1	0	0	0	1	1
Jack Henry & Associates Inc	0	1	1	1	0	1	1	1	1
Marqeta Inc	0	0	1	1	1	1	0	0	1
Mastercard Inc	1	1	0	1	1	1	1	1	1
NCR Atleos Corp	1	0	0	1	0	1	0	0	0
OLB Group Inc/The	0	0	0	1	0	0	1	1	0
POSaBIT Systems Corp	0	0	0	1	0	1	0	1	0
PayPal Holdings Inc	1	0	1	1	1	1	1	1	1
Paymentus Holdings Inc	1	1	0	1	0	0	0	0	0
Payoneer Global Inc	1	0	1	1	0	1	0	0	1
Paysign Inc	0	0	0	0	0	1	1	0	0
Priority Technology Holdings Inc	0	0	1	1	1	1	0	1	0
Remitly Global Inc	0	0	1	1	0	0	0	0	1
Repay Holdings Corp	0	0	0	1	0	0	0	0	0
Ryvyl Inc	1	0	1	1	0	1	0	0	0
Sezzle Inc	0	0	1	0	1	0	0	0	0
Shift4 Payments Inc	1	0	1	1	1	0	1	1	1
Toast Inc	0	0	0	1	1	1	0	1	0
Usio Inc	1	0	1	1	1	1	0	0	0
Visa Inc	0	1	0	1	1	1	1	0	1
WEX Inc	1	0	1	1	0	0	0	0	1
Western Union Co/The	1	0	1	1	0	0	0	1	1
XBP Global Holdings Inc	0	0	0	1	0	1	0	0	1

Notes: This table classifies payment firms according to the functions they provide. Closed-loop and open-loop operators manage private and open payment networks, respectively. Open-loop participants offer payment products that operate on existing open networks. Processors provide back-end payment services such as credential verification, authorization, and network communication. Inter-party services deliver payment-related services beyond the direct transfer of value between firms or individuals, while intra-party services provide services internal to a single firm or individual. Gateways supply front-end interfaces that enable digital payments, and physical POS providers manufacture or operate physical payment terminals. Cross-border firms coordinate payment correspondence and settlement across two or more national jurisdictions.

Table B.3: Firms classified by primary business function

Payment Networks ($\mathcal{G}^N$)	Cross-Border Payments ($\mathcal{G}^X$)	Other (Processing & Services)
Bank networks:	International Money Express Inc	Affirm Holdings Inc
American Express Inc	Payoneer Global Inc	Cass Information Systems Inc
FIS Inc	Remitly Global Inc	Change Financial Ltd
Jack Henry & Associates Inc	Western Union Co/The	Chime Financial Inc
Mastercard Inc		Corpay Inc
Visa		Flywire Corp
		Global Payments Inc
Consumer networks:		Marqeta Inc
Block Inc		Paymentus Holdings Inc
Cantaloupe Inc		Paysign Inc
Euronet Worldwide Inc		Priority Technology Holdings Inc
NCR Atleos Corp		Repay Holdings Corp
OLB Group Inc/The		Ryvyl Inc
PayPal Holdings Inc		Sezzle Inc
POSaBIT Systems Corp		Shift4 Payments Inc
Toast Inc		Usio Inc
		WEX Inc
		XBP Global Holdings Inc

Notes: This table shows payment firms classified by primary business function. Firms in the “Payment Networks” category are classified as being in $\mathcal{G}^N$. Firms in the cross-border payments category are classified as being in group $\mathcal{G}^X$. Firms in the “Other” category offer services such as payment processing, authentication, or card issuance. These firms do not primarily run their own networks, but instead add value to payment ecosystems run by other entities.

Table B.4: Firms engaged with crypto assets prior to the House passage of S.1582 on July 17, 2025

Firm	Examples of crypto-related activities	Source
Block Inc	Block offered Bitcoin via Cashapp among other crypto-related services.	Block.xyz
Canteloupe Inc	Cantaloupe Inc allowed cryptocurrency acceptance at POS terminals.	Payments Dive
Fidelity National Information	Fidelity partnered with Circle to offer stablecoin money movement for financial institutions.	FISGlobal.com
Fiserv Inc	Fiserv launched a stablecoin, FIUSD, for financial institutions.	Fiserv.com
Jack Henry & Associates Inc	Jack Henry & Associates partnered with NYDIG to offer bitcoin services to financial institutions.	JackHenry.com
Marqeta Inc	Marqeta offered crypto card issuing.	Marqeta.com
Mastercard Inc	Mastercard offered numerous crypto solutions to financial institutions.	Mastercard.com
NCR Atleos Corp	NCR acquired LibertyX to add bitcoin services at ATMs.	NCRAtleos.com
PayPal Holdings Inc	PayPal offered a USD denominated stablecoin, PayUSD.	Paypal.com
Ryvyl Inc	Ryvyl offered crypto treasury solutions for corporations.	Ryvyl.com
Shift4 Payments Inc	Shift4 offered crypto payment processing.	Shift4.com
Visa Inc	Visa piloted stablecoin settlement services.	Visa.com

Notes: This table shows payment firms that had engaged with crypto assets prior to 17 July 2025, along with examples of their crypto-related activities. These firms form our sub-group $\mathcal{G}^E$.

C Robustness

In this appendix, we show that our results are robust to alternative methodological choices, including considering alternative control groups and examining a broader set of legislative events. We also run placebo tests to confirm that our results are specific to legislative windows containing stablecoin news shocks.

Alternative control groups. A potential concern is that the control group—non-payments financial sector firms—might also be affected by the GENIUS Act, confounding our estimates. Figure C.1a shows that our results are robust to alternative control groups. First, we first exclude banks from our baseline control group. Second, we compare payment firms instead to all firms in the S&P 500 except those in the financial sector. In both cases, the coefficients remain very similar to those from our baseline specification, suggesting that any such effects of the GENIUS Act on the baseline control group are not substantial.

Additional controls. Another possible concern is that our results are driven by differences in other characteristics of the firms in the “treatment” and “control” groups, such as size or profitability, rather than by the payment firm versus non-payment firm distinction. To account for such possibilities, we supplement our baseline specification by allowing for differential trends by firm-level characteristics. We estimate:

$$\begin{aligned} \ln(P_{i,t}) = & \alpha_i + \alpha_t + \sum_{\substack{\tau=t^*-20 \\ \tau \neq t^*-1}}^{t^*+20} \gamma_\tau (\mathbf{1}_{\{t=\tau\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}}) \\ & + \sum_{\substack{\tau=t^*-20 \\ \tau \neq t^*-1}}^{t^*+20} \alpha'_\tau (\mathbf{1}_{\{t=\tau\}} \times \mathbf{X}_i) + \epsilon_{i,t} \end{aligned} \quad (6)$$

where $\mathbf{X}_i$ is a vector of ex-ante firm characteristics—specifically, market capitalization, revenue, net income, and return on assets. Intuitively, this specification controls for the possibility of differential trends across firms based on size or profitability.

Figure C.1b plots the estimated γ_τ coefficients from equation (6). Results remain very similar to those estimated using our baseline specification, providing reassurance that our results do indeed reflect a response of payment firms specifically, rather than reflecting other differences between the treatment and control firms.

Alternative event window widths. To check the sensitivity of our results to the definition of our event window, we repeat our baseline specification for window widths ranging from 4 to 48 trading hours. Estimates from longer windows can shed additional light on the persistence of the effect, but are also more likely to be contaminated by other shocks. Table C.1 shows that our baseline results are qualitatively and quantitatively similar with window widths between 4 and 24 trading hours. Even with a 48 trading-hour window, our market capitalization-weighted estimate is significantly negative at the 10% level, implying that the market capitalization-weighted returns of payment firms were persistently lower for more than three trading days after the event.

Pooling multiple legislative events. To confirm that our results are not specific to the House vote on S.1582, we also use a broader set of legislative events related to the passage of the bill. We examine all successful roll-call floor votes (formal votes in which all House or Senate members' votes are recorded) on the main text of the bill. Table C.2 lists these events. We identify all periods within a ten-hour window w around each vote, stacking them to create a new estimation sample. We convert all periods to event time and run the following specification at the 30-minute level:

$$\ln(P_{i,\tilde{t},w}) = \alpha_{i,w} + \alpha_{\tilde{t},w} + \sum_{\substack{\tau=-10 \\ \tau \neq -1}}^{10} \gamma_\tau (\mathbf{1}_{\{\tilde{t}=\tau\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}}) + \epsilon_{i,\tilde{t},w} \quad (7)$$

where event time $\tilde{t} \in \{-10, \dots, 10\}$ denotes time relative to the vote in each window, γ_τ represents the differential return of payment stocks at event time τ , and $\alpha_{i,w}$ and $\alpha_{\tilde{t},w}$ represent firm by window and event time by window fixed effects respectively. Intuitively, we create a panel that aligns five event windows and run our baseline specification across all events. We cluster standard errors at the firm times window (i, w) level.

Results based on specification (7) are shown in Figure C.1c and labeled as “Equal scaling”. The differential response of payment stocks is similar to our baseline estimate. This mitigates the concern that simultaneous, confounding shocks drive our baseline results, since it is highly unlikely that such shocks materialized simultaneously with all the other legislative events that we consider.

Specification (7) treats each event in the broader set equally when estimating the differential response of payment firms. Our prediction market data, however, suggest that different events had different impacts on the probability that investors assigned to U.S. stablecoin legislation occurring. We therefore run a second version of the pooled specification in which we scale the impact in each window w to have a magnitude comparable to our baseline specification:

$$\ln(P_{i,\tilde{t},w}) = \alpha_{i,w} + \alpha_{\tilde{t},w} + \sum_{\substack{\tau=-10 \\ \tau \neq -1}}^{10} \gamma_{\tau} \left(\mathbf{1}_{\{\tilde{t}=\tau\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}} \times \frac{\Delta p_w}{0.07} \right) + \epsilon_{i,\tilde{t},w} \quad (8)$$

where Δp_w is the size of the probability update that occurs between the first and second half of window w , and 0.07 is the size of this update for our baseline window (the decisive vote in the House of Representatives). Intuitively, the scale factor $\frac{\Delta p_w}{0.07}$ adjusts across windows to ensure that all windows are estimated using a shock equivalent to a seven percentage point increase in the prediction market-implied probability of U.S. stablecoin legislation passing in 2025. Results based on specification (8) are shown in Figure C.1c and labeled as “Scaling by probability update”. Once again, the estimated effect is of similar magnitude.

We can also run a difference-in-differences version of specification (8) (analogous to specification (3) in our baseline methodology) and weight by market capitalization:

$$\ln(P_{i,\tilde{t},w}) = \alpha_{i,w} + \alpha_{\tilde{t},w} + \gamma \left(\mathbf{1}_{\{\tilde{t} \geq 0\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}} \times \frac{\Delta p_w}{0.07} \right) + \epsilon_{i,\tilde{t},w} . \quad (9)$$

This produces an estimate of negative 1.6 percent—slightly larger in magnitude than our baseline estimate of negative 1.3 percent in Table 1—and hence an estimate for the full impact of the policy

of approximately 22.6 percent or \$375 billion, again slightly larger than our baseline estimate of \$300 billion.

Other crypto-related legislation. Another possible concern with our baseline specification is that another crypto-related bill, [H.R.3633](#) (the Digital Asset Market Clarity Act), also passed the House earlier on the same day, as part of the broader “Crypto Week” legislative push. There are several factors suggesting that this second vote is unlikely to drive our results. First, this bill had not previously passed the Senate, unlike S.1582, reducing the immediate implications of the second vote for markets.²² Second, H.R.3633 focuses on regulating the offer and sale of digital commodities such as bitcoin by the Securities and Exchange Commission and the Commodity Futures Trading Commission, rather than on the use of stablecoins in payments. As such, there is no evident reason why it should have a differential impact on payment firms relative to the rest of the financial sector. Third, the other events in our pooled specification above did not align with similar votes on H.R.3633, so this second bill could not explain those results.

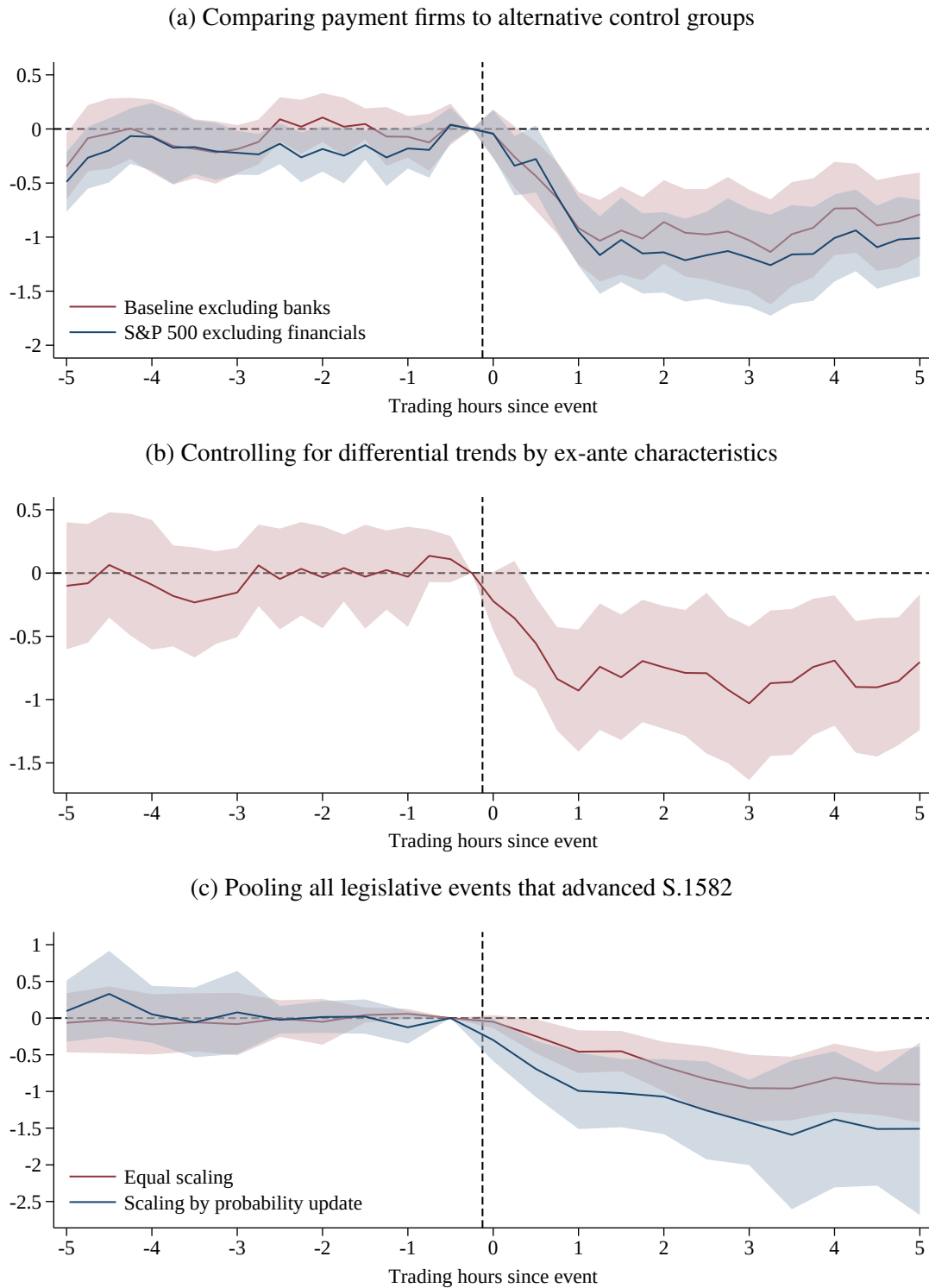
Nonetheless, in Figure [C.2a](#) we use one-minute-level data to examine the precise timing of the differential returns of payment firms that we estimate. We find that voting on H.R.3633 actually coincided with a small (but not statistically significant) *rise* in the relative returns of payment firms. Only after the voting on S.1582 did the significant negative differential returns of payment firms emerge. Indeed, when comparing the 20-minute intervals immediately following the decisive vote on each bill, we see no statistically significant effects for H.R.3633, but statistically significant negative effects for S.1582.

Placebo windows. We conduct placebo tests to confirm that our results are specific to legislative windows that contain stablecoin news shocks. We randomly select 1000 ten-hour windows (with replacement) from the set of all possible windows between 1 May 2025 (when S.1582 was introduced into Congress) and 18 July 2025 (when the bill was signed into law), excluding the windows that we use in our baseline and robustness regressions. For each of these 1000 windows,

²²Indeed, at the time of writing, this bill still has yet to pass the Senate and its future is [uncertain](#).

we then repeat our baseline specification and record the resulting coefficients γ_τ . Figure C.2b plots the results, which show that the average differential return of payment firms is centered on zero in these placebo windows. In other words, the returns of our treatment and control groups followed a similar path on average throughout the period that S.1582 was in Congress, when excluding the specific windows that we study. This supports our use of the control group as a counterfactual for the treatment group during our treatment windows, in line with a causal interpretation of our baseline results.

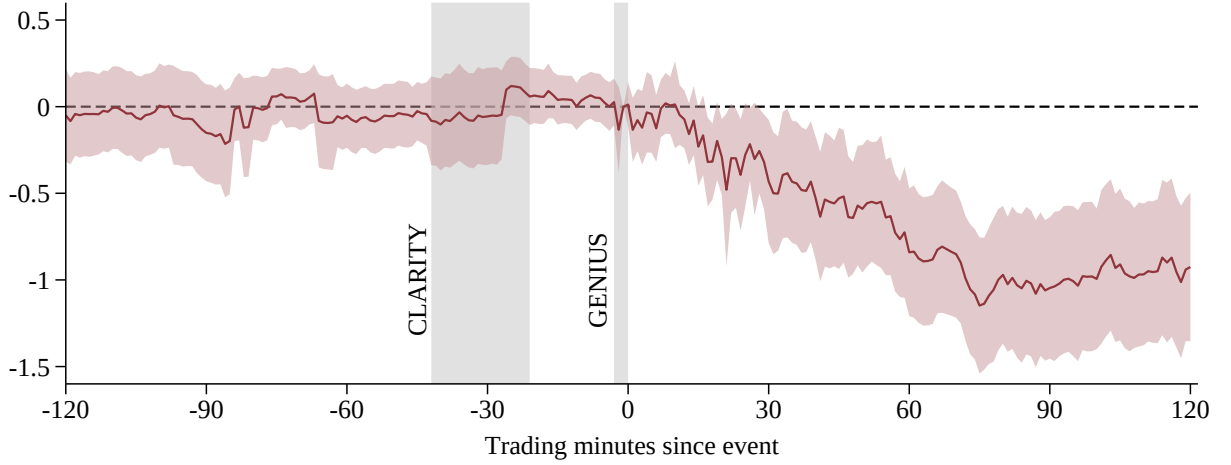
Figure C.1: Differential return of payment firms around vote window (percentage points)



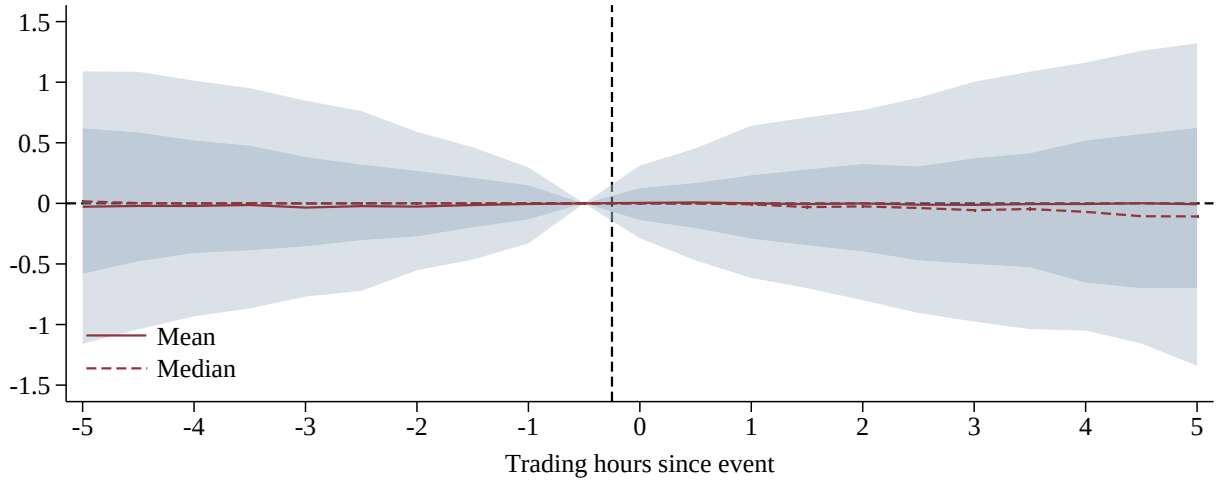
Notes: This figure shows the results from our robustness specifications, described in Appendix C. Panel (a) compares payment firms to alternative control groups, specifically (i) non-payment financial firms excluding banks and (ii) S&P 500 firms excluding financial firms. Panel (b) adds controls for differential trends by ex-ante firm characteristics, as in equation (6). Panel (c) shows our results when taking the average differential response across many legislative events using specifications (7) and (8).

Figure C.2: Differential return of payment firms (percentage points)

(a) Comparing the votes on H.R.3633 and S.1582



(b) Placebo windows



Notes: Panel (a) shows the differential return of payment firms when running an analogous specification to that in equation (2), at one-minute frequency for a four-hour window surrounding the decisive vote in the House of Representatives. The first shaded region indicates the interval between the first vote and the decisive vote on H.R.3633 (the CLARITY Act). The second shaded region indicates the interval between the first vote and the decisive vote on S.1582 (the GENIUS Act). Panel (b) plots the distribution of the differential returns of payment firms across 1000 randomly selected windows between 1 May and 18 July 2025, excluding the windows within which the legislative events that we study occurred. The solid (dashed) red lines plot the mean (median) coefficient estimated for each period in the window, and the light (dark) shaded regions indicate the 5th and 95th (25th and 75th) percentiles.

Table C.1: Average differential return of payment firms after the House passage of S.1582 (percentage points), using alternative event windows

	(1)	(2)	(3)	(4)	(5)
$\mathbf{1}_{\{t \geq t^{Pass}\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}}$ (unweighted)	-0.666*** (-3.07)	-0.727*** (-3.14)	-0.744*** (-3.15)	-0.707* (-1.67)	-1.435 (-1.27)
$\mathbf{1}_{\{t \geq t^{Pass}\}} \times \mathbf{1}_{\{i \in \mathcal{G}^P\}}$ (weighted)	-1.083** (-2.64)	-1.268*** (-3.56)	-1.253*** (-3.77)	-1.278*** (-3.08)	-1.024* (-1.73)
Firm FEs	✓	✓	✓	✓	✓
Period FEs	✓	✓	✓	✓	✓
Window width (hours)	4	8	12	24	48
Implied estimate (\$ billion)	-220	-292	-331	-470	-389
N (unweighted)	11,492	22,308	33,124	65,572	130,468
N (weighted)	11,543	22,407	33,271	65,863	131,047

Notes: t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered by firm and by period. This table repeats our baseline specification using event windows of different widths. $\mathcal{G}^P$ represents the set of all U.S.-listed payment firms. Coefficients in the first row reflect the average differential return of payments firms relative to other non-payments financial firms, in percentage points. Coefficients in the second row reflect the average differential return of payment firms relative to other non-payments financial firms, in percentage points, when observations are weighted by each firm’s pre-event market capitalization. Our baseline specification uses a 10-hour window. Columns (1) through (5) report results using windows with widths from 4 hours to 48 hours. The “Implied estimate (\$ billion)” row re-computes, for each window width, our estimate of the full impact on the aggregate payment sector market capitalization of the probability of U.S. stablecoin legislation in 2025 going from 0% to 100%, using the relevant weighted estimate and the corresponding change in prediction market-implied probabilities.

Table C.2: Successful roll-call floor votes on the main text of S.1582

Time of decisive vote	Chamber	Vote type	Description
May 19, 2025: 9:28pm	Senate	Cloture	The bill was introduced into the Senate on May 1, 2025. For debate to begin on a bill, a <i>motion to proceed</i> —a procedural vote to place the bill on the Senate’s agenda for consideration—must pass. The motion is typically brought by the majority leader. Before it can be voted on, there is often debate regarding whether and when to introduce the bill. This period of debate has no set duration as the Senate has no maximum duration for speeches or discussion. An action known as a <i>cloture</i> —a vote of three fifths of the chamber—can end the debate period, sending the action to a vote. On May 6, a motion to proceed on S.1582 was proposed by the majority leader. After a period of debate and agenda setting, a <i>cloture vote passed</i> on May 19. This ended debate on the motion, sending the motion to a formal vote.
May 21, 2025: 12:03pm	Senate	Motion to Proceed	The <i>motion to proceed</i> , as proposed on May 6, successfully passed in the Senate. This brought S.1582 onto the Senate floor, allowing members to begin formal debate and consideration of the bill.
June 12, 2025: 2:12pm	Senate	Cloture	A <i>cloture motion</i> can also be used to end the main period of debate in the Senate. If successful, it forces the Senate to vote on a bill’s passage directly. After nearly a month of debate, amendments, and alterations, a <i>cloture vote passed</i> regarding S.1582. This ended all debate and alterations, sending the bill to a vote. The vote was scheduled for five days later, June 17.
June 17, 2025: 5:25pm	Senate	Passage	The bill <i>passed the Senate</i> . Upon successful passage, the bill was then sent to the House of Representatives for consideration, which could then send the bill to committee for review or vote directly on the bill without further changes.
July 17, 2025: 3:32pm	House	Passage	When S.1582 entered the House of Representatives, the bill was <i>held at desk</i> —a procedure in which the bill is not referred to committee and instead sent directly to a debate and vote without any opportunity for alteration. This meant that, unlike the Senate proceedings, no procedural votes were necessary to begin debate or schedule votes. The bill was discussed and a vote scheduled for July 17. On July 17, there were multiple hours of contentious debate before the bill went to a formal vote. Ultimately, S.1582 <i>passed the House of Representatives</i> . After the vote, the bill was sent to the President’s desk and signed.

Notes: This table details all successful roll-call floor votes on the main text of the legislation throughout the passage of the GENIUS Act, including the time that the decisive vote occurred. The decisive vote is defined as the vote that put the total count over the required threshold necessary for passage. *Source:* [Congress.gov](https://www.congress.gov), with timestamps gathered manually from [C-SPAN](#) video recordings.

D Benchmarks

This appendix explains the events and sources that we use to benchmark our findings on the impact of stablecoins on payment firms.

Durbin Amendment. This amendment was attached to the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010 and capped debit card interchange fees for large banks.²³ This limited the potential for cross-subsidization from merchant fees to cardholder incentives, raising concerns that transaction volumes would fall. We use the estimate from [Berg, Keil, Martini, and Puri \(2024\)](#) that the Durbin Amendment reduced the returns of U.S. payment firms by 5.3% in the week that the regulation was announced and finalized.

Digital euro. [Berg, Keil, Martini, and Puri \(2024\)](#) argue that the introduction of a digital euro could have opposing effects on U.S. and European payment firms through a “strategic autonomy channel”. They present evidence that European regulators aim to use the introduction of a central bank digital currency to reduce European dependence on U.S. payments providers and, conversely, expand the role of European payments firms. We use their estimate that the shift from a 0% to a 100% probability of a digital euro being created reduced the weighted average returns of U.S. payment firm stocks by 11.6% and increased the weighted average returns of European payment firm stocks by 30.5%.

Credit Card Competition Act. The 2023 Senate bill [S.1838](#) (the Credit Card Competition Act) aimed to increase competition among credit card issuers by prohibiting large credit card issuers from restricting the number of networks on which credit card transactions can be processed. It [failed to progress](#) past the Senate Banking Committee, but was reintroduced into Congress (House bill [H.R.7035](#) and Senate bill [S.3623](#)) on January 13, 2026, shortly after President Trump [posted](#) his support for the Act. This [prompted](#) a decline in the stock prices of the largest credit card networks. To estimate the causal impact of this endorsement and the corresponding reintroduction

²³For further details, see [Kay, Manuszak, and Vojtech \(2018\)](#), [Wang \(2025\)](#) and [Mukharlyamov and Sarin \(2025\)](#).

of the bills, we follow a process analogous to our baseline methodology. Specifically, we estimate the following specification

$$\ln(P_{i,t}) = \alpha_i + \alpha_t + \gamma (\mathbf{1}_{\{t \geq t^*\}} \times \mathbf{1}_{\{i \in \mathcal{G}^N\}}) + \epsilon_{i,t} \quad (10)$$

for the ten trading hours surrounding the endorsement, for a sample containing (i) all incumbent payment networks $\mathcal{G}^N$ and (ii) our baseline control group (i.e., all non-payments financial sector firms). Weighting payment networks by ex-ante market capitalization, we find that their returns fell by 3.4% more than those of the control group.

RBI prohibition on new deposits into Paytm Payments Bank. One97 Communications Limited (hereafter, OCL) is a listed Indian financial technology firm offering a suite of services to consumers and merchants centered on its popular digital payments app, Paytm. Customer wallet balances are held at its affiliate, Paytm Payments Bank. On January 31, 2024, the Reserve Bank of India (RBI) ordered Paytm Payments Bank to stop accepting fresh deposits, citing persistent non-compliance with banking regulations and material supervisory concerns ([Reserve Bank of India, 2024](#)). Given the fierce competition between India’s digital payments platforms (for details, see [Copestake, Kirti, and Martinez Peria, 2025a](#)), this was a serious threat to OCL’s business model and indeed led to [a sharp increase in downloads](#) of alternatives to Paytm’s digital wallet.

We estimate the impact of this announcement on the stock returns of OCL using the following specification, analogous to our baseline methodology:

$$\ln(P_{i,t}) = \alpha_i + \alpha_t + \gamma (\mathbf{1}_{\{t \geq t^*\}} \times \mathbf{1}_{\{i = \text{OCL}\}}) + \epsilon_{i,t} \quad (11)$$

where $\mathbf{1}_{\{i = \text{OCL}\}}$ indicates OCL. Our control group is the set of all firms in the [NIFTY Financial Services Index](#), which (i) includes, for example, banks, asset managers, and insurers, analogous to the control group in our baseline regressions, and (ii) does not include OCL itself.

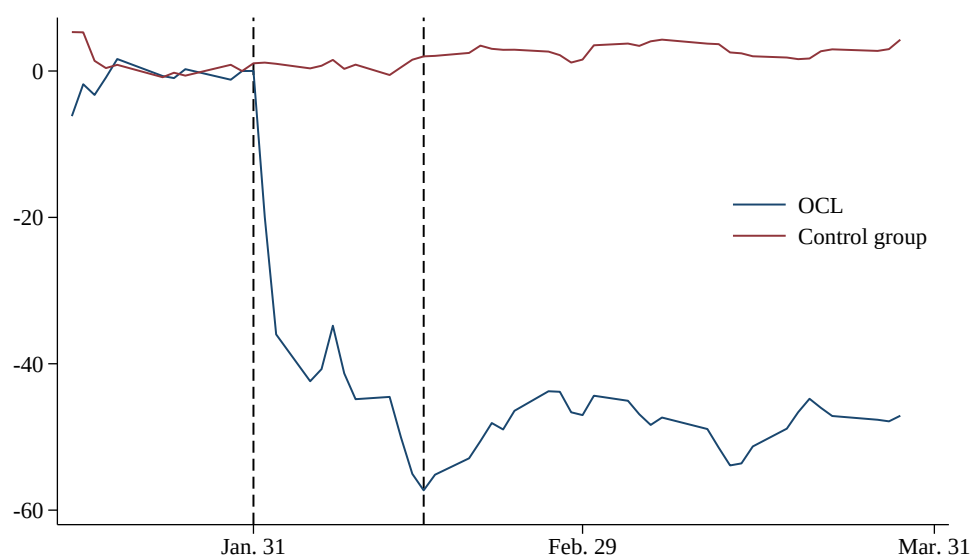
We bound the size of the effect by considering two different symmetrical estimation windows.

First, we calculate the differential returns using the average stock price on the day before and on the day after the announcement.²⁴ Second, we use a wider window of 15 days either side of the announcement, reflecting that OCL's stock price continued to decline through February 15 (see Figure D.1), stabilizing only on February 16 when OCL announced that it would substitute Paytm Payments Bank with a new account at Axis Bank and that Paytm's popular payments services (e.g., QR code-based payments and POS terminals) would continue to function for merchants (One 97 Communications Limited, 2024). We report estimates for both windows, reflecting the trade-off between excluding other shocks that could have affected OCL (which supports the narrower window) and capturing the full impact of the RBI's announcement (which supports the wider window). Weighting observations by market capitalization, we find differential returns of 11% using the narrower window and 48% using the wider window.

Suspension of major mobile payments platform IPO. One of the world's largest fintech firms, with more than one billion annual active users, planned an initial public offering (IPO) in 2020. The IPO was suspended at short notice and the firm was ultimately subjected to tighter regulation, such as higher capital requirements on its lending business. These developments led early investors (who had already invested prior to the planned IPO) to mark down the value of their stakes in the firm. We use estimates from Fidelity, which recorded a 51% reduction in the overall value of the firm after the IPO suspension. This reduction also aligns closely with public estimates by several analysts immediately after the IPO suspension.

²⁴We cannot examine intra-day changes since the RBI announcement does not contain an intra-day timestamp.

Figure D.1: Change in stock price of One97 Communications Ltd. and control group firms around RBI announcement (percent)



Notes: This figure shows the change in the stock price of One97 Communications Ltd. after the RBI's announcement on January 31, 2024. The figure also shows the corresponding change in the market capitalization-weighted average price of firms in the NIFTY Financial Services Index. Vertical lines show the date of RBI's initial announcement and OCL's response, as described in the text. Both price series are reported as a percent change relative to the period directly preceding the RBI's initial announcement. *Source:* Bloomberg.

E Additional descriptive evidence

This appendix presents descriptive evidence on actions taken by incumbent payment firms since the passage of the GENIUS Act. We show that the proportion of payment firms offering crypto-related products or services increased after the passage of the Act, as did the frequency with which stablecoins are mentioned in earnings and conference calls.

Engagement with crypto assets. In Appendix B, we define $\mathcal{G}^E$ as the set of payment firms that had issued a crypto asset or offered services utilizing an existing crypto asset prior to the decisive House vote on S.1582 on July 17, 2025. We now extend this measure to other points in time, adding three more dates: the start of 2025, the day the bill entered the U.S. Senate (May 1), and the end of 2025. The red line in Figure E.1 plots the share of payment firms that had engaged with crypto assets by each of these dates. The share of crypto-engaged firms was flat in 2025 prior to the bill entering Congress, increased slightly as the bill progressed, and increased more quickly throughout the remainder of the year after the bill passed.

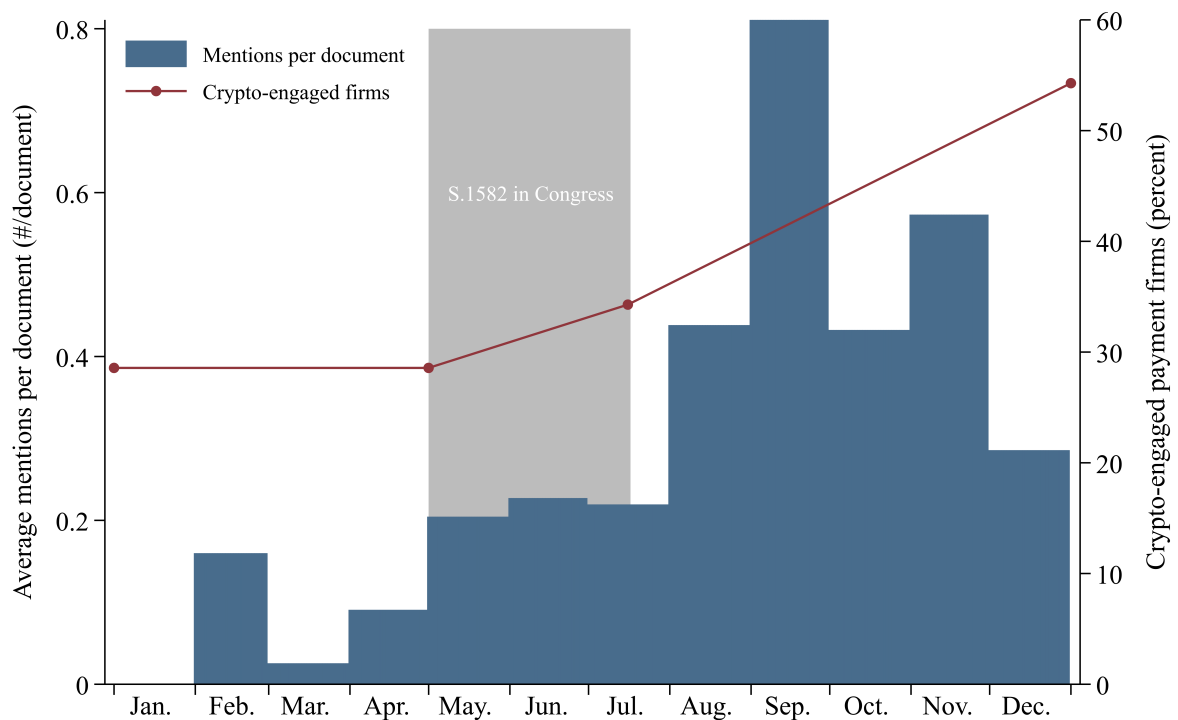
Earnings calls. We examine how often payment firms discuss stablecoins in their communications with investors. To do this, we gather more than 10,000 earnings and conference call transcripts, covering 97% of the payment firms in our sample, and count the number of times that stablecoins are mentioned in each document.²⁵ The vertical bars in Figure E.1 show the average number of mentions per document per month. We see that stablecoins were rarely mentioned prior to the introduction of S.1582, then discussed more frequently while it was being debated in Congress. After the GENIUS Act became law, mentions increased substantially, to a peak of nearly one mention per call in September 2025.

Table E.1 shows examples of the ways in which stablecoins were discussed by different types of firm. The first two quotes, from a cross-border correspondent, illustrate how some firms' expressed views on the utility of stablecoins changed quickly after the passage of the GENIUS

²⁵Specifically, we look for sentences containing the word “stablecoin” or the names of prominent stablecoins (e.g. “USDT”, “USDC”).

Act. Indeed, the prominent cross-border payments firm Western Union [announced in October 2025](#) that it would launch its own stablecoin.

Figure E.1: Firm engagement with crypto assets around passage of the GENIUS Act



Notes: This figure shows descriptive evidence on engagement with crypto assets by payment firms in our sample surrounding the passage of the GENIUS Act. The red line plots the number of firms that had issued a crypto asset, or offered services utilizing an existing crypto asset, prior to the specified date. The vertical bars show the average number of mentions of stablecoins per document from a corpus of payment firms' earnings and conference calls. The shaded region indicates the period between S.1582 being introduced into the Senate and it being passed by the House of Representatives. *Source:* LSEG.

Table E.1: Excerpts from payment firms' earnings calls

Firm type	Quote
Cross-border correspondence	May 2025: "It's not yet clear what the value proposition is because it's complex and expensive to take from fiat currency, move into a stablecoin, move the stablecoin across geographies, translate the stablecoin back into local currency, the transaction costs and the logistics of getting high-quality exchanges on both ends to do that, enough volume to be able to do it in and out of the local currency is, again, operationally challenging and expensive." July 2025: "Stablecoin enabled solutions have the potential to reduce friction and flow, enabling faster, more efficient movement of value between our global network and local payout partners." "... with the passage of the GENIUS Act, we are actively revising our position with a clear regulatory framework and guaranteed redemption value."
Card network	July 2025: "Beyond capital markets use cases, we see product market fit for stablecoins in two important areas: one, in emerging markets where the local fiat currency is volatile and/or where consumers do not have easy or affordable access to US dollars; and two, in cross-border money movement, both B2B payments and consumer." "We are supportive of the GENIUS Act and we believe that it marks a key milestone on the path to regulatory clarity for stablecoins."
Bank payment services	August 2025: "Last week, we announced we're expanding ... capabilities to include digital assets through our partnership with Circle Internet Group, enabling banks to transact in USDC for both domestic and cross-border payments."
Payments processing	September 2025: "And the clear value here is while real-time payments can allow you to move money instantaneously and super cost-effectively, which is a core part of sort of the stablecoin value proposition or blockchain, what stablecoin or blockchain does is sort of the always-on, 24/7 element." October 2025: "Certainly, there are use cases that I think stablecoins can help solve, particularly around cross-border business-to-business payments, how settlements and payouts are made to different customers."

Notes: This table shows excerpts from payment firm earnings calls that mention stablecoins. *Source:* LSEG.



PUBLICATIONS

Stablecoins and the Future of Payments

Working Paper No. WP/2026/052